

Simplex III: Two-Phase Simplex, Geometry, and Efficiency

How do we start simplex, interpret its basis changes, and understand its computational cost?

3.5 · Finding an initial feasible basis

3.6 · Column geometry

3.7 · Computational efficiency

Lecture 7 · Bertsimas–Tsitsiklis §§3.5–3.7

Use the pivot, reduced-cost, and anticycling rules developed in Simplex I and II.

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How Do We Find the First Feasible Basis?

An anticycling rule tells simplex how to continue without repeating bases. We still need a place to start.

The missing input

A **basic feasible solution**, together with its basis matrix and constraint rows.

The two-phase idea

Phase I: find a feasible basis or prove infeasibility.

Phase II: optimize the original objective from that basis.

An Easy Start: Nonnegative Slacks

Suppose A has m rows and n columns, and the constraints are

$$Ax \leq b, \quad x \geq 0, \quad b \geq 0.$$

Add a vector $s \in \mathbb{R}^m$ of nonnegative **slack variables**:

$$Ax + s = b, \quad x, s \geq 0.$$

$$x = 0, \quad s = b, \quad B = I_m.$$

The slack columns form the identity matrix I_m ; their values are nonnegative. This is a basic feasible solution.

Prepare the Equality Constraints

Now consider the general standard-form linear program:

$$\min f(x) = c^\top x \quad \text{subject to } Ax = b, \quad x \geq 0.$$

Here $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, and c is the original cost vector.

If a right-hand side $b_i < 0$, multiply **that entire equality** by -1 .

$$A_{i:}x = b_i \quad \Longleftrightarrow \quad (-A_{i:})x = -b_i.$$

After these equivalent row changes, assume $b \geq 0$. This alone does not supply an original-variable basis.

Artificial Variables Give an Initial Basis

Introduce $y \in \mathbb{R}^m$, a vector of **artificial variables**. Define the auxiliary objective $w = \sum_{i=1}^m y_i$.

$$\begin{array}{ll} \min & w = \mathbf{1}^\top y \\ \text{subject to} & Ax + y = b, \\ & x \geq 0, \quad y \geq 0. \end{array}$$

$$x = 0, \quad y = b, \quad B = I_m.$$

The artificial columns supply a feasible basis immediately. The vector $\mathbf{1}$ contains m ones.

The original objective $c^\top x$ waits until Phase II. Phase I asks whether the artificial variables can all become zero.

Why Zero Phase I Cost Means Feasibility

If $Ax = b$, $x \geq 0$, then $(x, 0)$ satisfies the auxiliary constraints and has $w = 0$.

Conversely, if an auxiliary feasible point has $w = 0$,

$$y \geq 0, \quad \sum_i y_i = 0 \quad \implies \quad y = 0.$$

Thus $Ax + y = b$ becomes $Ax = b$.

Original LP feasible $\iff w^* = 0.$

The symbol w^* denotes the optimal Phase I objective value. Nonnegativity of each artificial variable is essential.

A Positive Phase I Optimum Proves Infeasibility

Phase I starts feasible and its objective satisfies $w \geq 0$, so it cannot be unbounded below.

$$w^* > 0 \implies \text{the original LP is infeasible.}$$

An original feasible point would give auxiliary cost zero, contradicting optimality of $w^* > 0$.

A **current** positive Phase I cost is inconclusive. Continue until the auxiliary optimum is certified. A feasible auxiliary point of cost zero already reaches the lower bound.

Example 3.8: Start Without an Obvious Basis

We will carry this complete model through both phases:

$$\begin{array}{ll}\min & f = x_1 + x_2 + x_3 \\ \text{subject to} & x_1 + 2x_2 + 3x_3 = 3, \\ & x_1 - 2x_2 - 6x_3 = -2, \\ & 4x_2 + 9x_3 = 5, \\ & 3x_3 + x_4 = 1, \quad x_1, x_2, x_3, x_4 \geq 0.\end{array}$$

Reverse the second equality to make its right-hand side positive. Keep the original costs $c = (1, 1, 1, 0)$.

Example 3.8: Build the Auxiliary LP

In this example, name the four artificial variables x_5, x_6, x_7, x_8 .

$$\begin{array}{ll}\min & w = x_5 + x_6 + x_7 + x_8 \\ \text{subject to} & x_1 + 2x_2 + 3x_3 + x_5 = 3, \\ & -x_1 + 2x_2 + 6x_3 + x_6 = 2, \\ & 4x_2 + 9x_3 + x_7 = 5, \\ & 3x_3 + x_4 + x_8 = 1, \quad x_1, \dots, x_8 \geq 0.\end{array}$$

Original variables $x_1, \dots, x_4$ have zero Phase I cost; every artificial variable has cost one.

Example 3.8: The Artificial Basis Costs 11

$$x_1 = x_2 = x_3 = x_4 = 0, \quad (x_5, x_6, x_7, x_8) = (3, 2, 5, 1).$$

The basis columns are the four artificial unit columns:

$$B = [A_5 \ A_6 \ A_7 \ A_8] = I_4, \quad w = 3 + 2 + 5 + 1 = 11.$$

Here $A_5, \dots, A_8$ denote columns of the **augmented** constraint matrix. The original matrix has only columns $A_1, \dots, A_4$.

Our task is to reduce w to zero while preserving the augmented equalities and nonnegativity.

Read the Objective Row Before We Pivot

Let $M = [A \mid I_4]$, $v = (x_1, \dots, x_8)$, and $q = (0, 0, 0, 0, 1, 1, 1, 1)$. The auxiliary objective is $w = q^\top v$.

$$w = w_0 + \bar{q}^\top v, \quad \bar{q}^\top = q^\top - q_{\mathcal{B}}^\top B^{-1} M.$$

Our tableau convention is

$$\begin{array}{c|c} -w_0 & \bar{q}^\top \\ \hline B^{-1}b & B^{-1}M \end{array}.$$

RHS means right-hand side. Its top entry is **minus** the current objective; entries below it are basic values.

R_0 names the objective row.

Construct the Phase I Objective Row

Initially $B = I_4$ and $q_B = \mathbf{1}$. Eliminate the basic artificial coefficients from the raw objective row.

$$\begin{aligned}\bar{q}^\top &= [\mathbf{0}^\top \mid \mathbf{1}^\top] - \mathbf{1}^\top [A \mid I_4] \\ &= [-\mathbf{1}^\top A \mid \mathbf{0}^\top].\end{aligned}$$

The four original column sums are 0, 8, 21, 1. Therefore

$$\bar{q} = (0, -8, -21, -1, 0, 0, 0, 0).$$

Subtract the four constraint rows from the raw objective row:

$$R_0 = [-11 \mid 0, -8, -21, -1, 0, 0, 0, 0].$$

Example 3.8: The Initial Tableau

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	−11	0	−8	−21	−1	0	0	0	0
x_5	3	1	2	3	0	1	0	0	0
x_6	2	−1	2	6	0	0	1	0	0
x_7	5	0	4	9	0	0	0	1	0
x_8	1	0	0	3	1	0	0	0	1

The upper-left entry is −11, so $w = 11$. The basic artificial columns form the identity.

We choose eligible entering variables in the reference order x_4, x_3, x_2, x_1 . Break leaving-variable ties by smaller variable index.

First choose x_4 : its reduced cost is −1, and its column is already a unit column.

Pivot 1: x_4 Enters and x_8 Leaves

Increase $x_4 = \theta$. Its only nonzero constraint coefficient is one in the x_8 row.

$$x_8 = 1 - \theta \geq 0 \quad \implies \quad \theta^* = 1.$$

Replace basic x_8 by x_4 . Their columns are identical, so the basis matrix remains I_4 .

$$w_{\text{new}} = 11 + (-1)(1) = 10.$$

Only the objective row changes numerically:

$$R_0 \leftarrow R_0 + R_4.$$

The fourth basic-variable label changes from x_8 to x_4 .

After Pivot 1: The Cost Is 10

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	−10	0	−8	−18	0	0	0	0	1
x_5	3	1	2	3	0	1	0	0	0
x_6	2	−1	2	6	0	0	1	0	0
x_7	5	0	4	9	0	0	0	1	0
x_4	1	0	0	3	1	0	0	0	1

The basis is (x_5, x_6, x_7, x_4) , with values $(3, 2, 5, 1)$.

Next choose x_3 , whose reduced cost is $−18$. Its tableau column is $(3, 6, 9, 3)^\top$.

Pivot 2: A Tie Sends x_4 Out of the Basis

For entering x_3 , divide each basic value by its positive entering-column coefficient:

Basic variable	Value	Coefficient	Step bound
x_5	3	3	1
x_6	2	6	1/3
x_7	5	9	5/9
x_4	1	3	1/3

The minimum is $\theta^* = 1/3$. The tie is between x_4 and x_6 ; choose x_4 .

$$w_{\text{new}} = 10 - 18(1/3) = 4.$$

After Pivot 2: A Basic Artificial Variable Is Zero

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	−4	0	−8	0	6	0	0	0	7
x_5	2	1	2	0	−1	1	0	0	−1
x_6	0	−1	2	0	−2	0	1	0	−2
x_7	2	0	4	0	−3	0	0	1	−3
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$	0	0	0	$\frac{1}{3}$

The basis is (x_5, x_6, x_7, x_3) . Artificial variable x_6 is still basic, but its value is now zero.

The remaining negative reduced cost is $\bar{q}_2 = -8$, so choose x_2 next.

Pivot 3: The Step Is Zero

The entering x_2 column is $(2, 2, 4, 0)^\top$, in basic order (x_5, x_6, x_7, x_3) .

$$\theta^* = \min \left\{ \frac{2}{2}, \frac{0}{2}, \frac{2}{4} \right\} = 0.$$

Basic x_6 leaves and x_2 enters at zero. The basis changes while the point and objective stay fixed:

$$w_{\text{new}} = 4 + (-8)(0) = 4.$$

Normalize the pivot row by dividing by 2; eliminate x_2 from every other row, including R_0 .

After Pivot 3: Same Point, Different Basis

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	−4	−4	0	0	−2	0	4	0	−1
x_5	2	2	0	0	1	1	−1	0	1
x_2	0	$−\frac{1}{2}$	1	0	−1	0	$\frac{1}{2}$	0	−1
x_7	2	2	0	0	1	0	−2	1	1
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$	0	0	0	$\frac{1}{3}$

The basis is (x_5, x_2, x_7, x_3) , with values $(2, 0, 2, 1/3)$. The objective remains $w = 4$.

Choose x_1 , whose reduced cost is $−4$. Its entering column is $(2, −1/2, 2, 0)^T$.

Pivot 4: Remove Another Artificial Variable

Increase x_1 . Only the positive coefficients in its tableau column give upper bounds.

$$\theta^* = \min \left\{ \frac{x_5}{2}, \frac{x_7}{2} \right\} = \min\{1, 1\} = 1.$$

The $-1/2$ coefficient makes basic x_2 increase; the zero coefficient leaves x_3 unchanged.

The leaving tie is between x_5 and x_7 . Choose x_5 , then pivot on the coefficient 2.

$$w_{\text{new}} = 4 - 4(1) = 0.$$

Phase I Reaches Zero, with x_7 Still Basic

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	0	0	0	0	0	2	2	0	1
x_1	1	1	0	0	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$
x_2	$\frac{1}{2}$	0	1	0	$-\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	0	$-\frac{3}{4}$
x_7	0	0	0	0	0	-1	-1	1	0
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$	0	0	0	$\frac{1}{3}$

All artificial values are zero, so the original point is $(x_1, x_2, x_3, x_4) = (1, 1/2, 1/3, 0)$.

The basis still includes artificial $x_7 = 0$. Inspect its row before starting Phase II.

A Feasible Point Is Not Yet the Starting Basis

Suppose Phase I ends at (x^*, y^*) with $w = 0$. Then $y^* = 0$, and x^* is feasible for the original LP.

An artificial variable can still be **basic at value zero**. The auxiliary basic feasible solution is then degenerate.

What Phase II needs

A basis made entirely of original columns, its constraint rows, and reduced costs for the **original** objective.

If the Final Basis Already Uses Original Columns

Suppose the final auxiliary basis B consists entirely of columns of A . Keep its feasible basic values $B^{-1}b$.

Delete artificial columns. The original constraint rows remain

$$[B^{-1}b \mid B^{-1}A].$$

Using the original costs c , rebuild the objective information:

$$f_0 = c_{\mathcal{B}}^{\top} B^{-1}b, \quad \bar{c}^{\top} = c^{\top} - c_{\mathcal{B}}^{\top} B^{-1}A.$$

Here $c_{\mathcal{B}}$ lists costs in the same order as the columns of B . Continue simplex with this original objective.

Why Cleanup Is Possible When A Has Full Row Rank

Suppose the final auxiliary basis contains $k < m$ original columns. Reorder the basis so these are $A_{B(1)}, \dots, A_{B(k)}$.

They are linearly independent because they are part of a basis.

If $\text{rank}(A) = m$, the columns of A span $\mathbb{R}^m$. Extend those k independent columns to m independent original columns:

$$\widetilde{B} = [A_{B(1)} \ \cdots \ A_{B(k)} \ A_{B(k+1)} \ \cdots \ A_{B(m)}].$$

The symbols $B(i)$ name column indices; B and $\widetilde{B}$ are basis matrices.

The Extended Basis Represents the Same Point

Every positive original component of x^* was already basic. Original variables outside those k columns were nonbasic and therefore zero.

The added original columns receive value zero. With all other nonbasic values zero,

$$\tilde{B}x_{\tilde{\mathcal{B}}}^* = b, \quad x_{\tilde{\mathcal{B}}}^* \geq 0.$$

Nonsingularity makes these the basic values $\tilde{B}^{-1}b$. Thus x^* is unchanged.

If $\text{rank}(A) < m$, no such m -column basis exists. We must identify and remove redundant equality constraints.

Inspect the Row of a Zero Artificial Variable

Let row ℓ belong to an artificial basic variable with value zero. Write $H = B^{-1}A$ and $h = B^{-1}b$; H_{ij} is row i , column j .

Value	x_1	...	x_j	...	x_n	Artificial column
h_i	H_{i1}	...	H_{ij}	...	H_{in}	0
$h_\ell = 0$	$H_{\ell 1}$	...	$H_{\ell j}$	...	$H_{\ell n}$	1
h_r	H_{r1}	...	H_{rj}	...	H_{rn}	0

Rows $i, r \neq \ell$ stand for the other constraint rows. The basic artificial column is the unit vector e_ℓ .

Search only the original columns: is some $H_{\ell j} \neq 0$? A nonzero entry permits a basis exchange; an entirely zero row reveals a redundancy.

Case 1: A Nonzero Entry Supplies a New Column

Keep the original basic columns first, so their positions are $1, \dots, k$ and the artificial row satisfies $\ell > k$. The vector e_i has a one in position i , zeros elsewhere.

$$B^{-1}A_{B(i)} = e_i \quad (i = 1, \dots, k).$$

Every linear combination of these vectors has zero ℓ th component.

$$H_{\ell j} = (B^{-1}A_j)_\ell \neq 0 \quad \implies \quad B^{-1}A_j \notin \text{span}\{e_1, \dots, e_k\}.$$

Because B is invertible, A_j is independent of the original columns already in the basis.

Pivot at Zero: Even a Negative Entry Works

Choose a nonzero original entry $p = H_{\ell j}$. Let R_i denote the entire i th tableau row.

$$R_\ell \leftarrow R_\ell / p, \quad R_i \leftarrow R_i - H_{ij} R_\ell \quad (i \neq \ell).$$

The entering column becomes e_ℓ . Original variable x_j enters; the artificial variable leaves.

Since $h_\ell = 0$, scaling this row and adding it to other rows changes no basic value. The same feasible point remains, with one fewer basic artificial variable.

The pivot may be negative: this cleanup exchanges basis columns at a zero step.

Case 2: Every Original Entry in the Row Is Zero

Suppose $H_{\ell j} = 0$ for every original column $j = 1, \dots, n$.

Value	All original columns	Artificial column
0	0 ... 0	1

Let $g^\top = e_\ell^\top B^{-1}$, the ℓ th row of the inverse basis matrix.

$$g \neq 0, \qquad g^\top A = e_\ell^\top B^{-1} A = 0^\top.$$

A row of an invertible matrix cannot be zero. Thus g gives a nontrivial linear dependence among the rows of A .

Why the Zero Row Can Be Removed

Phase I has reached zero, so an original feasible point x^* exists.

$$g^\top b = g^\top Ax^* = 0.$$

The same combination of right-hand sides is therefore also zero.

$$g^\top Ax = g^\top b \iff 0 = 0.$$

After artificial variables are set to zero, this transformed tableau row imposes no restriction. Delete that redundant row and its artificial basic variable.

Repeat: pivot out a zero artificial variable when possible; otherwise delete its redundant row. Finish with an original-variable basis for the remaining independent constraints.

The x_7 Row Identifies an Exact Redundancy

The x_7 tableau row is

$$[0 \mid 0, 0, 0, 0 \mid -1, -1, 1, 0].$$

Every original-column entry is zero. The artificial block identifies $g = (-1, -1, 1, 0)$.

$$-A_{1:} - A_{2:} + A_{3:} = 0, \quad -3 - 2 + 5 = 0.$$

For the normalized model, row 3 equals row 1 plus row 2.

Delete this zero transformed row and the basic artificial variable x_7 . The remaining independent equations have a three-column basis.

Remove the Redundant Tableau Row

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	0	0	0	0	0	2	2	0	1
x_1	1	1	0	0	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$
x_2	$\frac{1}{2}$	0	1	0	$-\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	0	$-\frac{3}{4}$
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$	0	0	0	$\frac{1}{3}$

The remaining basic variables are x_1, x_2, x_3 . Their nonnegative values are unchanged.

No artificial variable is now basic. We can delete all four artificial columns, including nonbasic x_7 .

Remove Artificial Columns, Keep the Constraint Rows

Basic	RHS	x_1	x_2	x_3	x_4
R_0	0	0	0	0	0
x_1	1	1	0	0	$\frac{1}{2}$
x_2	$\frac{1}{2}$	0	1	0	$-\frac{3}{4}$
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$

The zero objective row still belongs to **Phase I**. It says nothing about optimality for $f = x_1 + x_2 + x_3$.

Use the original costs $c = (1, 1, 1, 0)$. In basic order (x_1, x_2, x_3) , $c_{\mathcal{B}} = (1, 1, 1)$.

Rebuild the Original Objective Row

The current original objective is $f_0 = 1 + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}$.

The basic reduced costs are zero. For x_4 , the transformed column is $(1/2, -3/4, 1/3)^\top$:

$$\bar{c}_4 = 0 - \left(\frac{1}{2} - \frac{3}{4} + \frac{1}{3} \right) = -\frac{1}{12}.$$

$$R_0 = \left[-\frac{11}{6} \mid 0, 0, 0, -\frac{1}{12} \right].$$

Feasibility survives cleanup. Reduced costs change because the objective changes.

Phase II Begins from a Feasible Original Basis

Basic	RHS	x_1	x_2	x_3	x_4
R_0	$-\frac{11}{6}$	0	0	0	$-\frac{1}{12}$
x_1	1	1	0	0	$\frac{1}{2}$
x_2	$\frac{1}{2}$	0	1	0	$-\frac{3}{4}$
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$

Try the next pivot

Which variable can enter? Which basic variable limits the step? Compute the next point and original objective before continuing.

The following slides give the complete solution to this exercise.

Phase II: x_4 Enters and x_3 Leaves

Since $\bar{c}_4 = -1/12$, let $x_4 = \theta$. The basic values change as

$$\begin{aligned}x_1 &= 1 - \theta/2, \\x_2 &= 1/2 + 3\theta/4, \\x_3 &= 1/3 - \theta/3.\end{aligned}$$

$$\theta^* = \min \left\{ \frac{1}{1/2}, \frac{1/3}{1/3} \right\} = 1.$$

Basic x_3 leaves. Basic x_2 increases and gives no upper bound.

$$f_{\text{new}} = \frac{11}{6} - \frac{1}{12} = \frac{7}{4}.$$

Example 3.8: The Original Optimum Is 7/4

Basic	RHS	x_1	x_2	x_3	x_4
R_0	$-\frac{7}{4}$	0	0	$\frac{1}{4}$	0
x_1	$\frac{1}{2}$	1	0	$-\frac{3}{2}$	0
x_2	$\frac{5}{4}$	0	1	$\frac{9}{4}$	0
x_4	1	0	0	3	1

$$x^* = (1/2, 5/4, 0, 1), \qquad f(x^*) = 7/4.$$

The sole nonbasic reduced cost is $\bar{c}_3 = 1/4 > 0$. The feasible basis is optimal for the original minimization problem.

Check the original four equalities directly. The redundant equality still holds even though we no longer store it.

Use Existing Unit Columns When Possible

In Example 3.8, x_4 appears only in the fourth equation, with coefficient one. Its column already supplies the fourth unit column.

Start with $(x_5, x_6, x_7, x_4) = (3, 2, 5, 1)$, omit artificial x_8 , and skip the first pivot.

More generally, an original variable occurring in only one normalized row with coefficient $\alpha > 0$ can be basic there at value $b_i/\alpha \geq 0$. Artificial columns are needed only for the remaining rows.

Keep the rows and chosen starting columns independent; the initial Phase I objective counts only artificial variables actually introduced.

Phase I: Establish Feasibility

- 1. Normalize:** multiply equality rows by -1 as needed so $b \geq 0$.
- 2. Initialize:** keep suitable original starting columns, introduce any needed artificial variables, and minimize their sum using simplex with an anticycling rule.
- 3. Read the optimum:** if $w^* > 0$, stop: the original LP is infeasible. If $w^* = 0$, all artificial values are zero and an original feasible point has been found.

Phase I: Turn the Feasible Point into a Basis

While an artificial variable is basic at zero, inspect its tableau row ℓ over original columns:

A nonzero original entry exists

Pivot on any $(B^{-1}A_j)_\ell \neq 0$. Original x_j enters; the artificial basic variable leaves. The point stays fixed.

All original entries are zero

The row is a redundant equality at zero Phase I cost. Delete it and its artificial basic variable.

Repeat until all basic variables are original. Discard the remaining artificial columns.

Phase II: Restore and Optimize the Original Cost

1. **Keep the basis** produced by Phase I after cleanup. It is feasible for the remaining independent original equalities.
2. **Keep the constraint rows** after removing artificial columns; discard the Phase I objective row.
3. **Reprice:** compute the current original objective f_0 and $\bar{c}_j = c_j - c_{\mathcal{B}}^{\top} B^{-1} A_j$ using the original costs.
4. **Continue simplex** with an anticycling rule until an optimal basis or an improving feasible ray is obtained.

Two Phases Handle Every Standard-Form Outcome

What happens?	Where it is handled
The original LP is infeasible	Phase I optimum $w^* > 0$.
Feasible equalities are dependent	Phase I cleanup removes redundant rows.
Original objective is unbounded below	Phase II finds an improving feasible ray.
A finite optimum exists	Phase II terminates at an optimal basis.

An anticycling rule supplies the termination guarantee, including degenerate cases.

We now know how to start and finish. Next, interpret these basis exchanges geometrically using the columns of the constraint matrix.

Column geometry: use the variables as mixture weights

The same basis calculations have a second geometric interpretation.

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & Ax = b, \quad \sum_{j=1}^n x_j = 1, \quad x \geq 0. \end{array}$$

The added **convexity constraint** makes the variables nonnegative weights that sum to one. Here A has m rows.

A bounded feasible LP can be reformulated this way. Self-study: derive such a reformulation by rescaling bounded nonnegative variables and adding one slack weight.

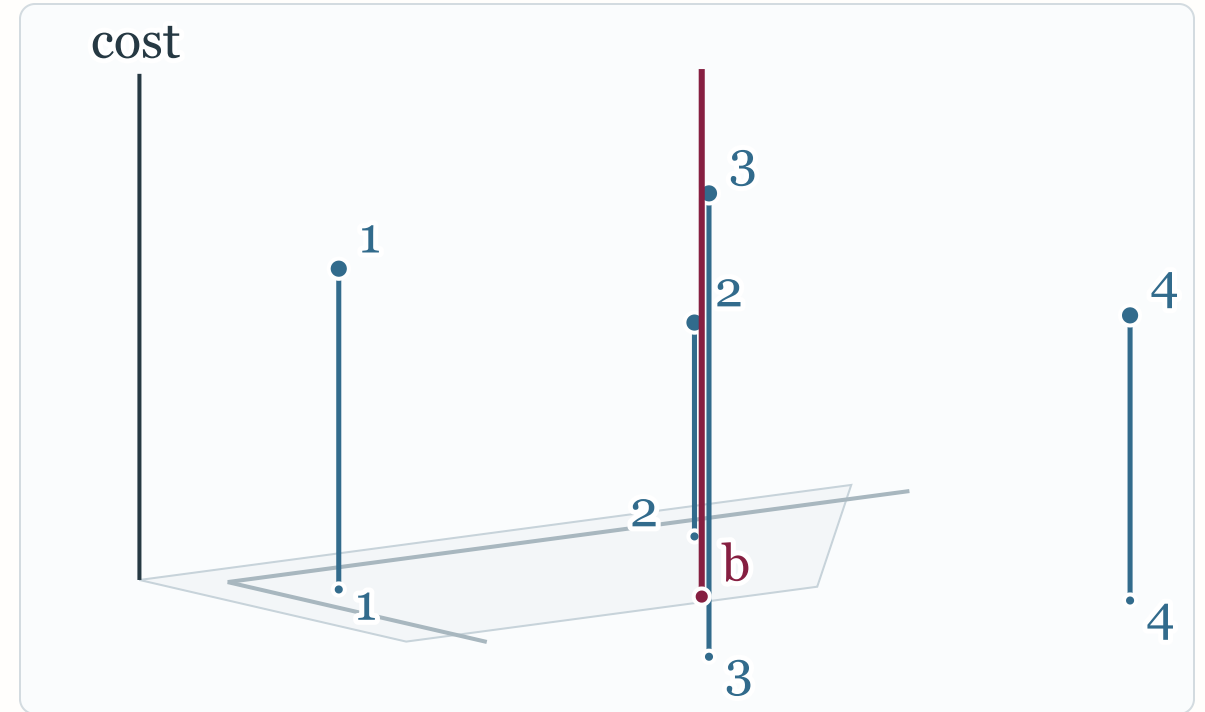
Lift each column to its objective height

Set $z = c^\top x$ and attach cost c_j to column A_j :

$$P_j = \begin{pmatrix} A_j \\ c_j \end{pmatrix}, \quad \sum_j x_j P_j = \begin{pmatrix} b \\ z \end{pmatrix}.$$

For $m = 2$, columns live in the horizontal plane. Their costs provide the third coordinate. The vertical line passes through b .

Each label j identifies column A_j at the floor and lifted point P_j above it.



A concrete realization of the five-point picture

We will use these constructed coordinates throughout the 3D example. A point name also labels its mixture weight.

Point	B	C	D	E	F
A_{1j}	1/2	3	0	2	0
A_{2j}	1/2	0	3	1/2	0
c_j	7	4	4	1	4

$$\min \sum_j c_j x_j \quad \text{s.t.} \quad \sum_j A_j x_j = (1, 1)^\top, \quad \sum_j x_j = 1, \quad x_j \geq 0.$$

The target horizontal coordinate stays fixed at $b = (1, 1)$.

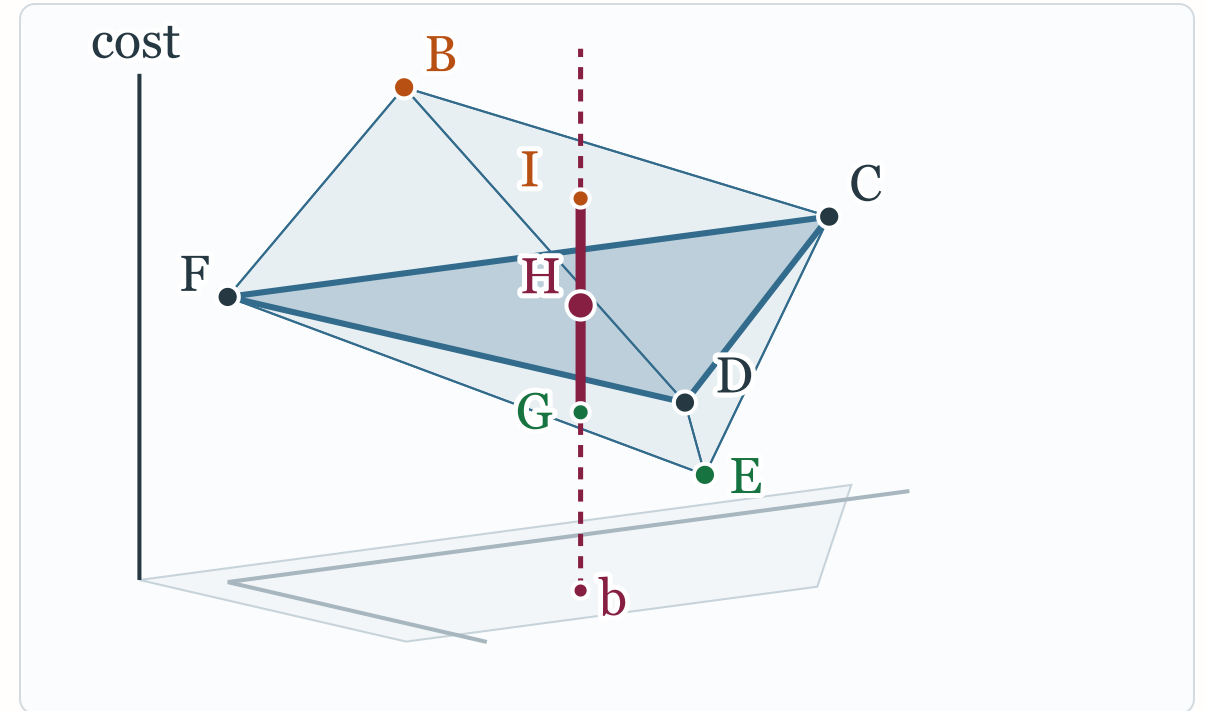
Feasibility is an intersection with a vertical line

A feasible weight vector places (b, z) inside the convex hull of the lifted columns.

The **requirement line** consists of all points (b, z) with the same horizontal coordinate b .

The thick vertical segment contains all attainable costs. Its lowest point, G, is optimal.

H and I are feasible too, but they sit higher.



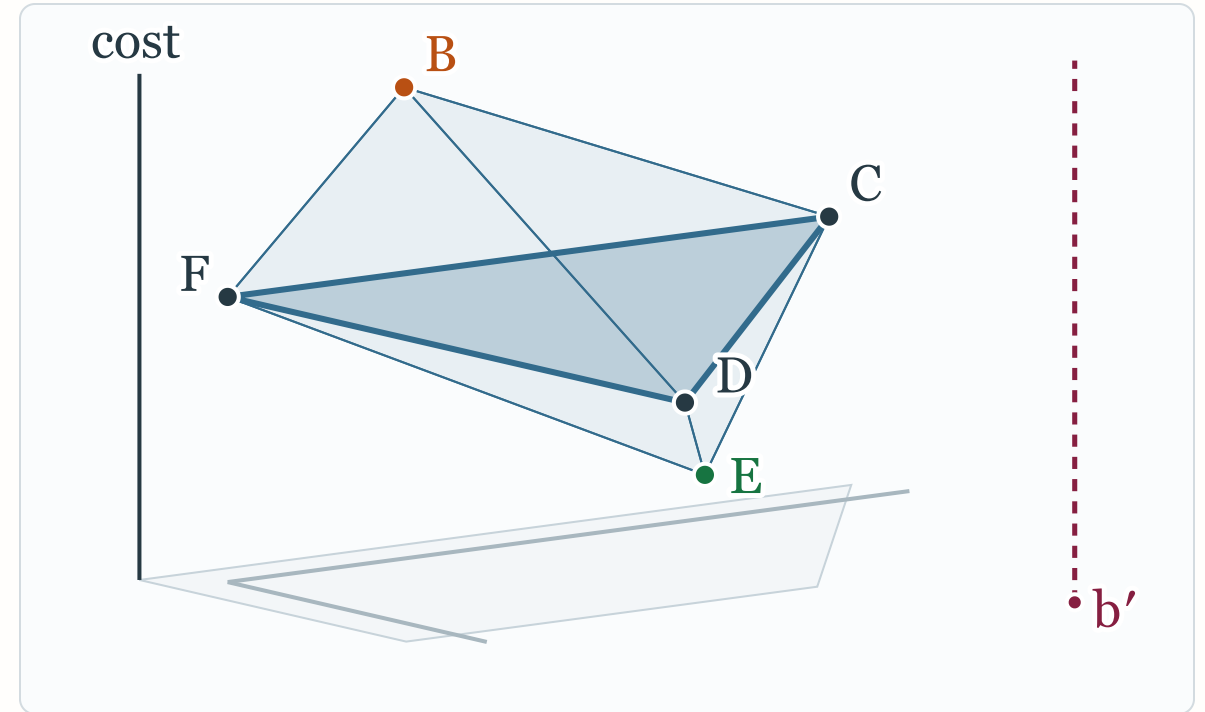
If the line misses the hull, no mixture is feasible

Keep the same five lifted columns, but change the requirement to $b' = (2.4, 2.4)$.

Every column satisfies $A_{1j} + A_{2j} \leq 3$.
Therefore every convex combination satisfies the same inequality.

$$b'_1 + b'_2 = 4.8 > 3.$$

The new requirement line never meets the hull.
The LP is infeasible, regardless of its objective.



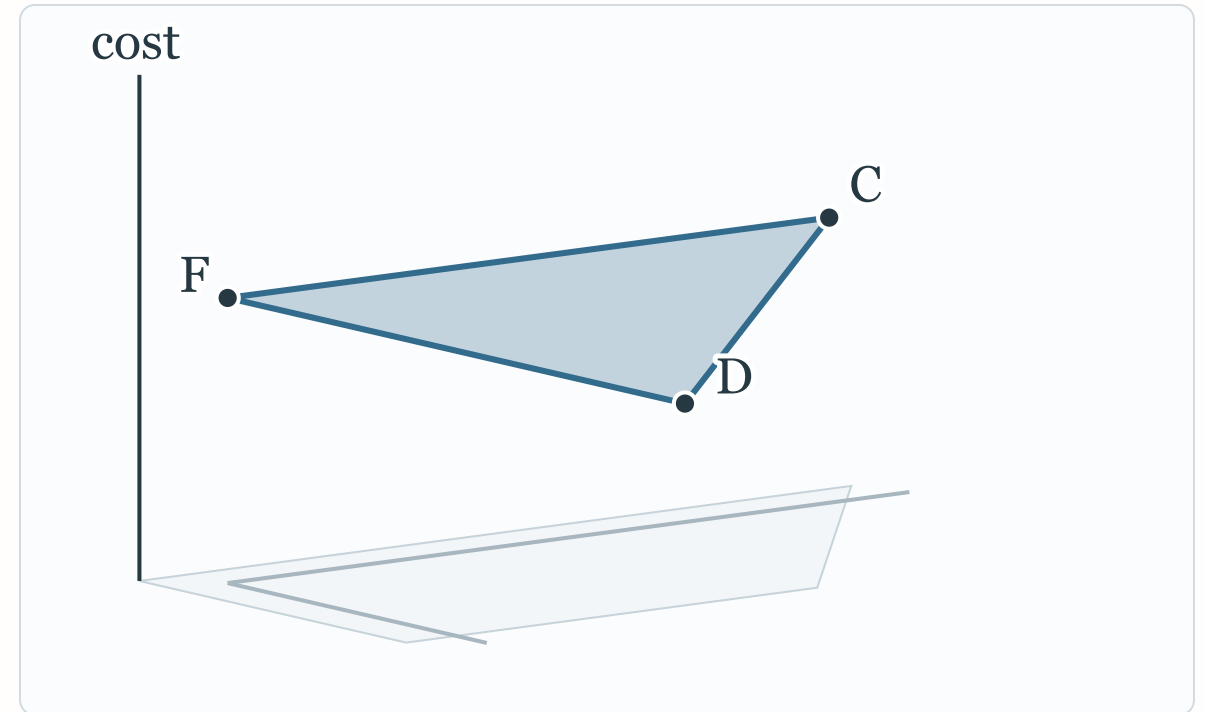
Affine independence gives a nonflat simplex

Points $v_1, \dots, v_{k+1} \in \mathbb{R}^n$ are **affinely independent** when

$$v_1 - v_{k+1}, \dots, v_k - v_{k+1}$$

are linearly independent. Necessarily $k \leq n$.

Their convex hull is a **k -dimensional simplex**.
Three noncollinear points give a triangle: CDF is a 2-simplex.



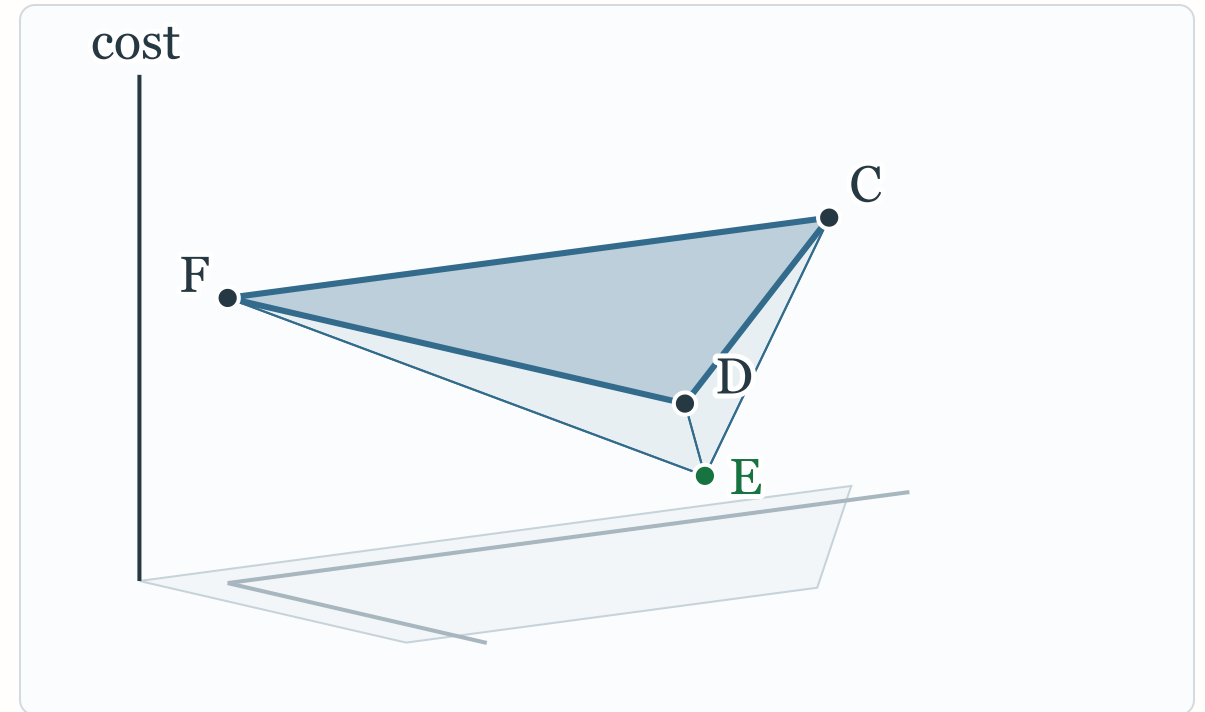
Four noncoplanar points give a tetrahedron

Four points are either coplanar or affinely independent.

In the independent case their convex hull is a 3-simplex: a tetrahedron, with four triangular faces.

C, D and F lie in one plane. E lies below it, so C, D, E and F are not coplanar.

Self-check: three differences from F span three dimensions. Which new direction does E contribute?



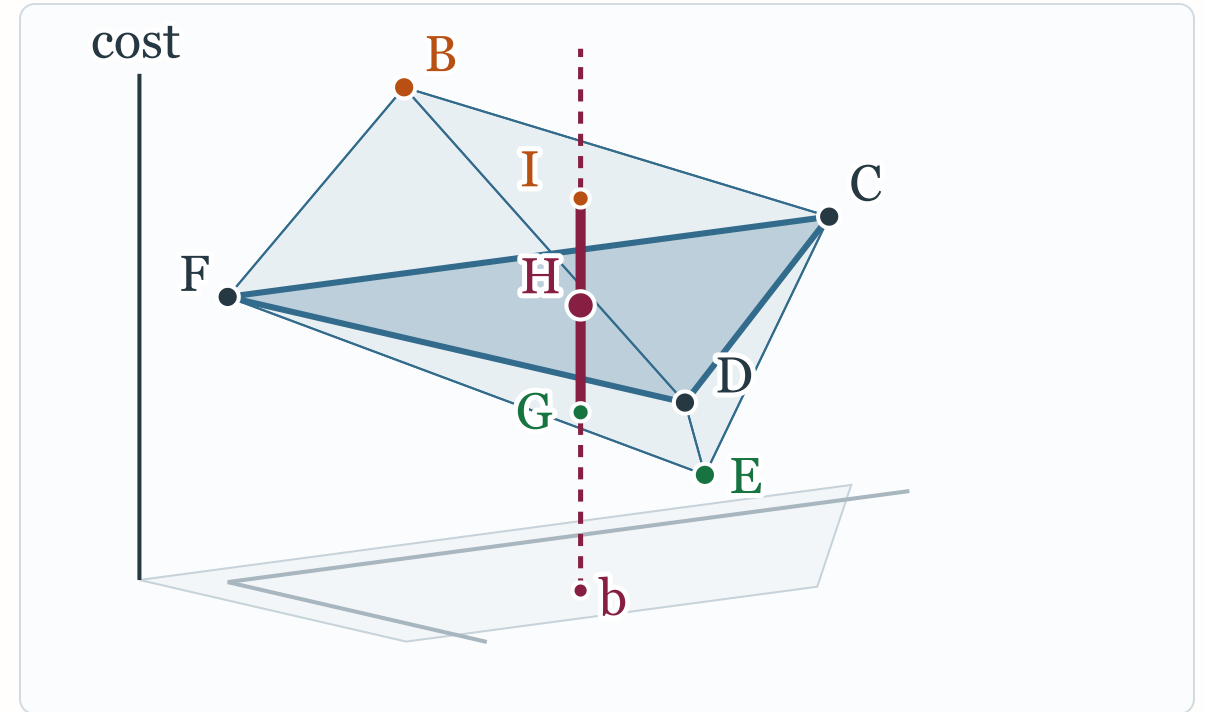
A basis selects three lifted points

Including convexity, the constraint matrix is

$$\tilde{A} = \begin{pmatrix} A \\ \mathbf{1}^\top \end{pmatrix}, \quad \tilde{b} = \begin{pmatrix} b \\ 1 \end{pmatrix}.$$

Assume $\tilde{A}$ has full row rank. A basis selects $m + 1$ independent columns $(A_j, 1)$.

Here $m = 2$: choose C, D, F. Their lifted versions are the **basic points**. B and E are nonbasic.

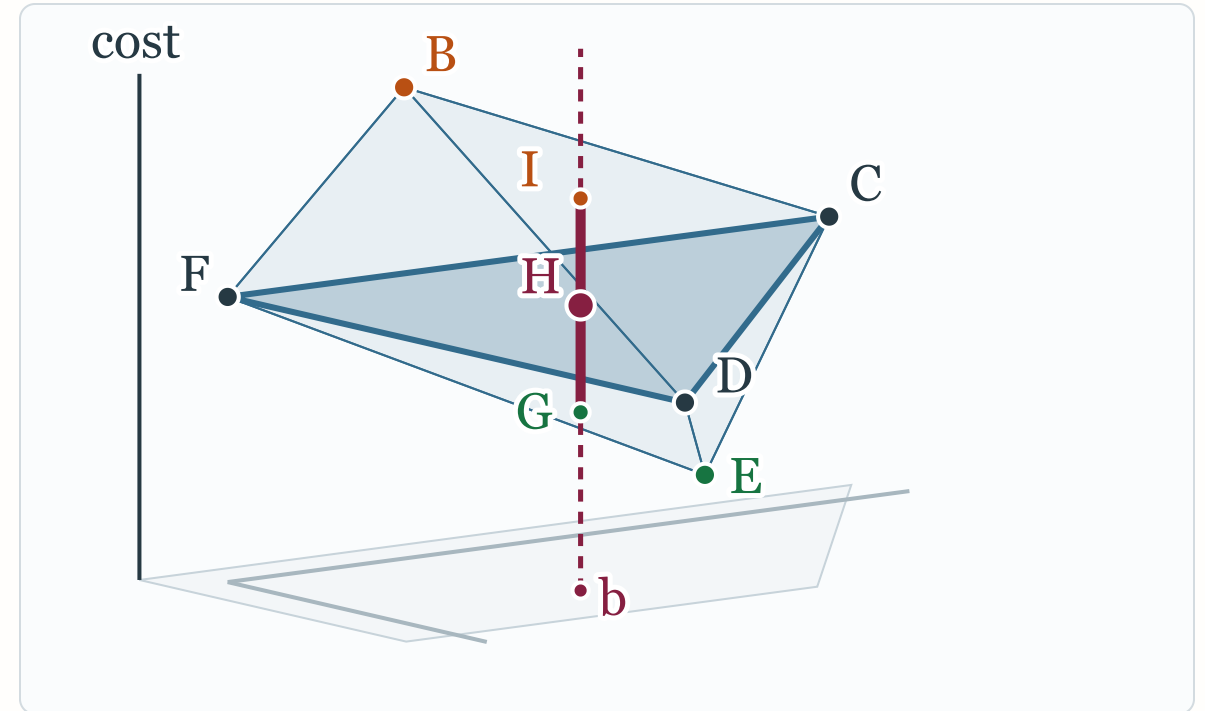


The selected points form the basic simplex

The basic columns $(A_j, 1)$ are independent, so their horizontal points A_j are affinely independent.

Lifting them to (A_j, c_j) preserves that affine independence: a relation between lifted differences would also be a relation between horizontal differences.

Their convex hull is the basic simplex. Our three basic points form the shaded triangle CDF.



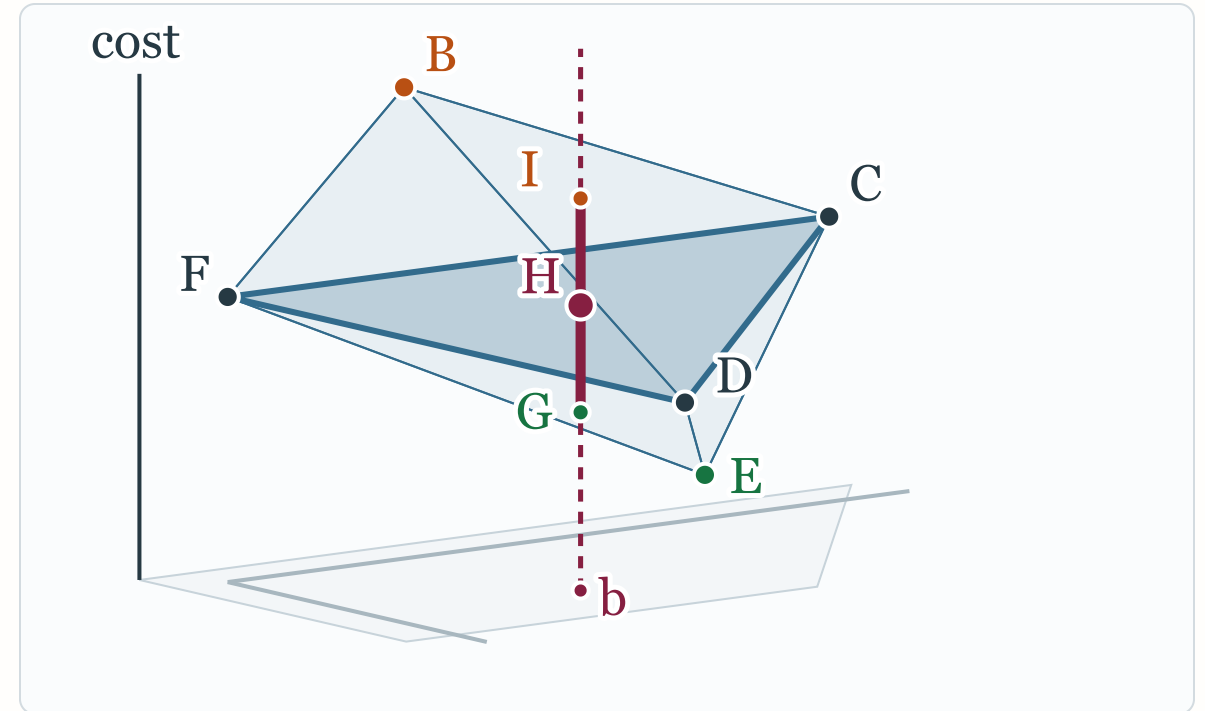
The intersection H determines the basic weights

Intersect the requirement line with triangle CDF.

$$H = \frac{1}{3}C + \frac{1}{3}D + \frac{1}{3}F = (1, 1, 4).$$

Thus $x_C = x_D = x_F = 1/3$, while $x_B = x_E = 0$. These are the entries of the **basic feasible solution**.

The point H is a lifted mixture. Its height 4 is the objective value; its coordinates are not the five decision variables.



A feasible basic mixture need not be optimal

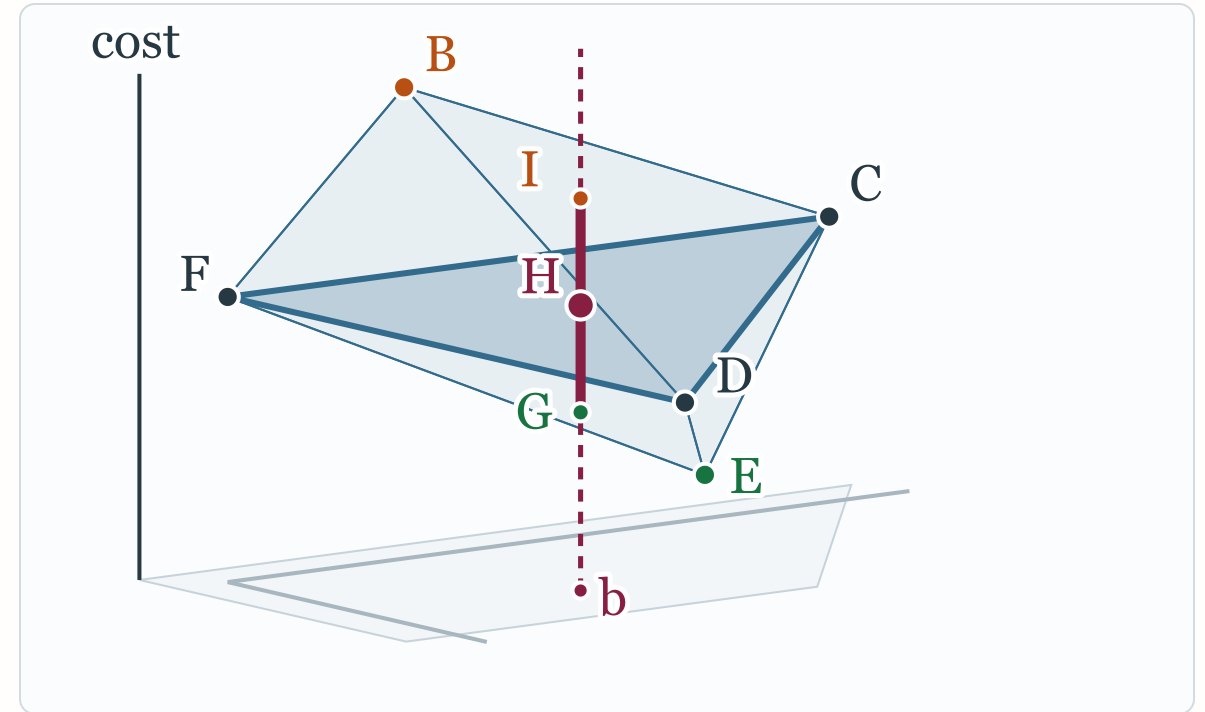
The same requirement line meets several possible basic triangles.

$$H = (1, 1, 4),$$

$$G = (1, 1, 5/2).$$

H uses CDF. The lower point G uses DEF. A change of basis can move the intersection downward while the requirement stays fixed.

The task is to choose a new basic point and an old basic point to remove.

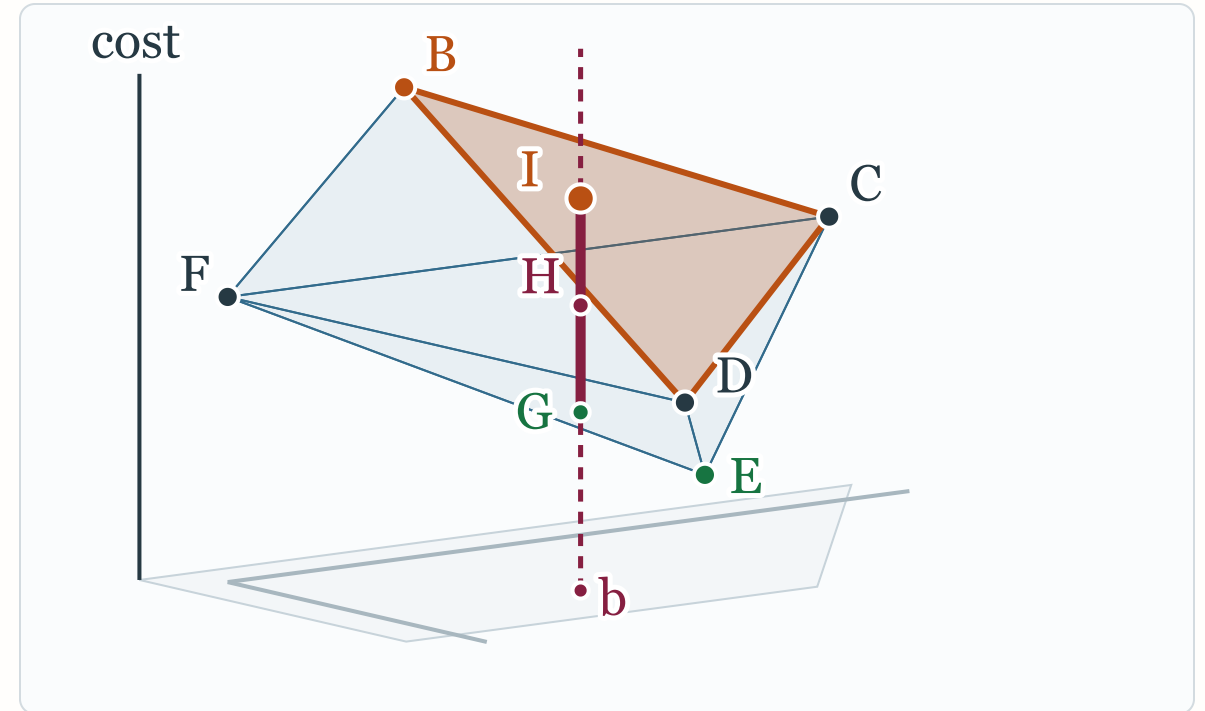


Candidate B: replace F and move upward to I

Starting with CDF, let B enter and F leave. The new basic simplex is BCD.

$$I = \frac{1}{2}B + \frac{1}{4}C + \frac{1}{4}D = (1, 1, 11/2).$$

The weights are feasible, but cost rises from 4 to $11/2$. This is a valid basis exchange with the wrong objective direction for minimization.

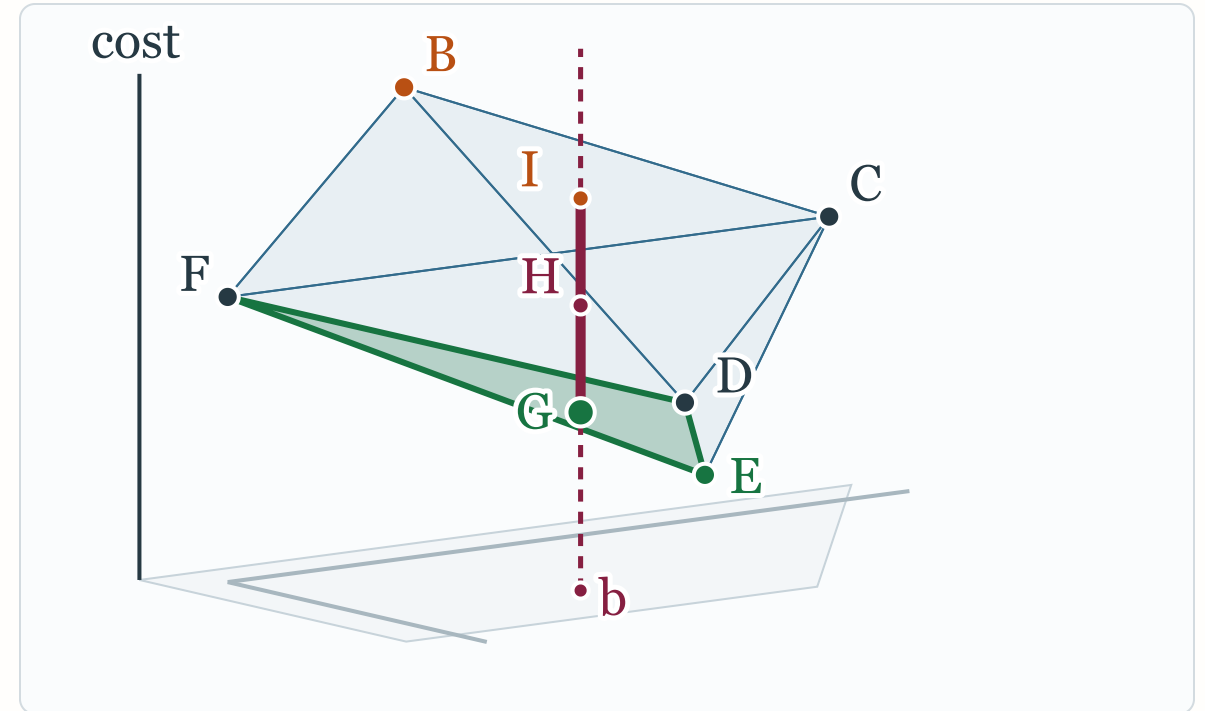


Candidate E: replace C and move downward to G

Instead, let E enter and C leave. The new basic simplex is DEF.

$$G = \frac{1}{4}D + \frac{1}{2}E + \frac{1}{4}F = (1, 1, 5/2).$$

Cost falls from 4 to $5/2$. The next question is how to predict the better entering point before completing the exchange.



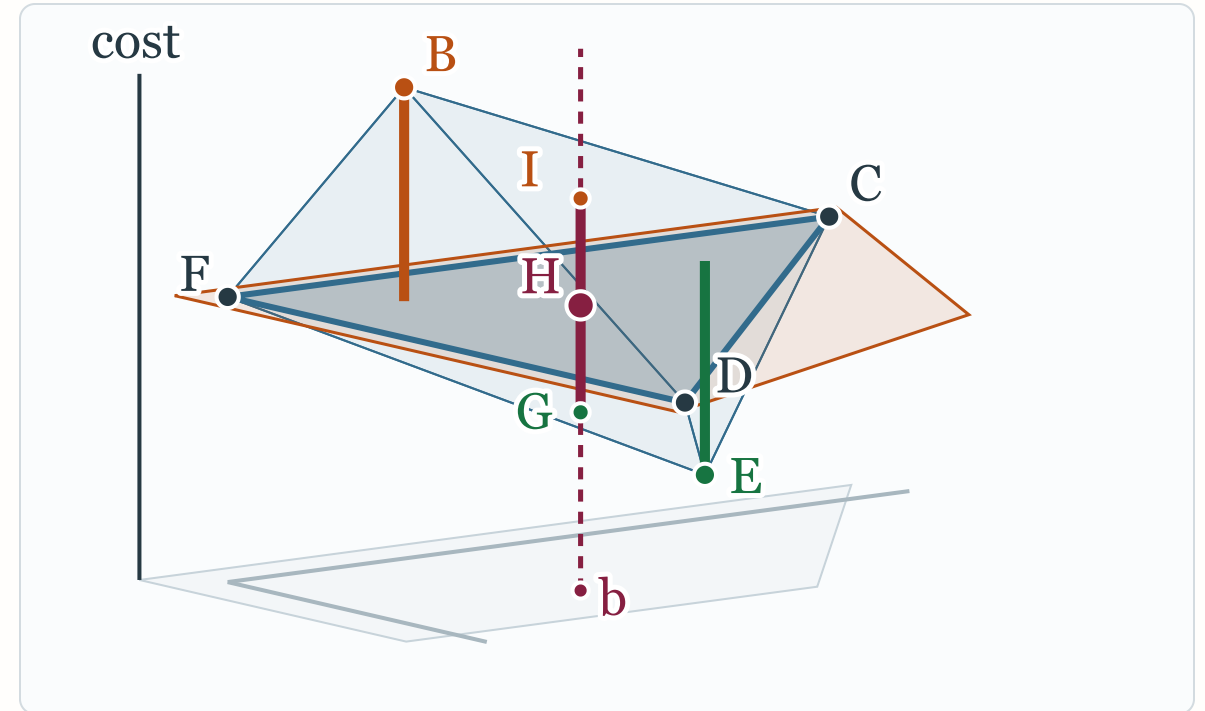
The dual plane predicts the objective slope

Extend the plane through the basic points C, D and F. This is the current dual plane.

$$z = 4.$$

E is below this plane; B is above it. Increasing the weight on E can decrease cost, while increasing the weight on B increases cost.

Strict improvement also needs a positive feasible step. A degenerate pivot may change the basis without changing the cost.



Reduced cost is a signed vertical gap

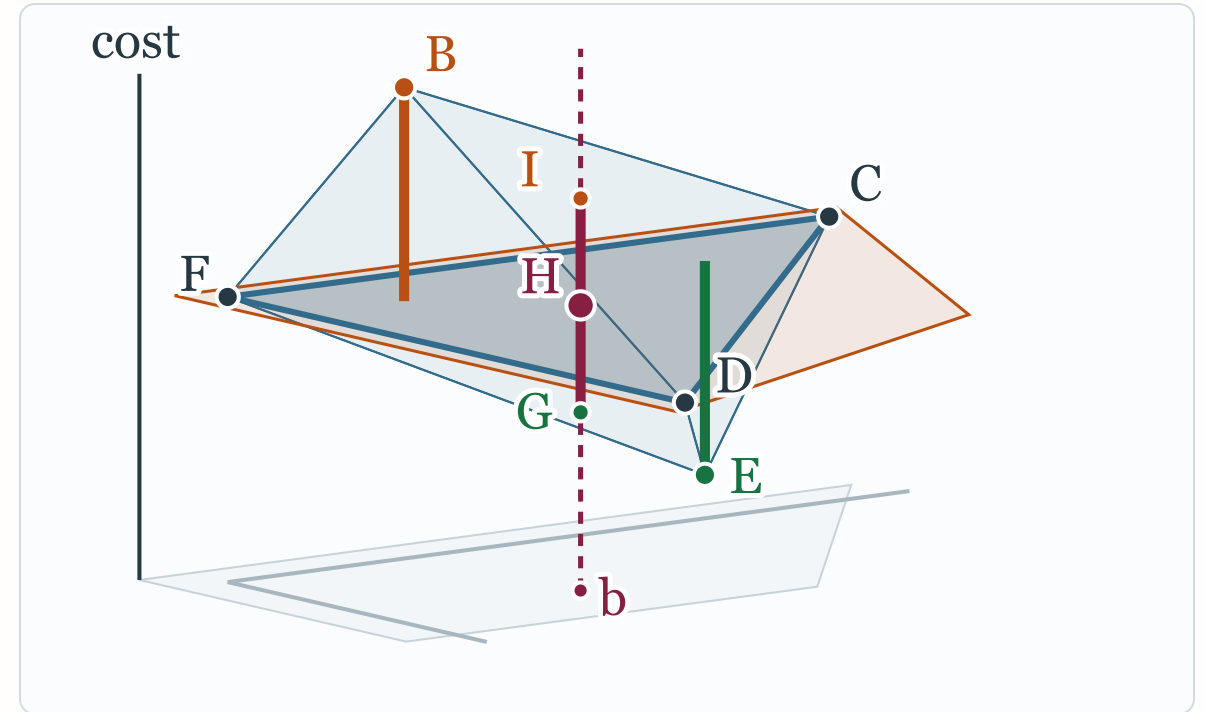
Write the dual plane as $z = \pi^\top a + \alpha$. It agrees with each basic point's cost.

$$\tilde{B}^\top \begin{pmatrix} \pi \\ \alpha \end{pmatrix} = c_B.$$

$$\bar{c}_j = c_j - (\pi^\top A_j + \alpha).$$

Subtract plane height from point height. Below the plane means a negative gap.

For CDF, $\pi = (0, 0)$, $\alpha = 4$: $\bar{c}_E = -3$, $\bar{c}_B = 3$.

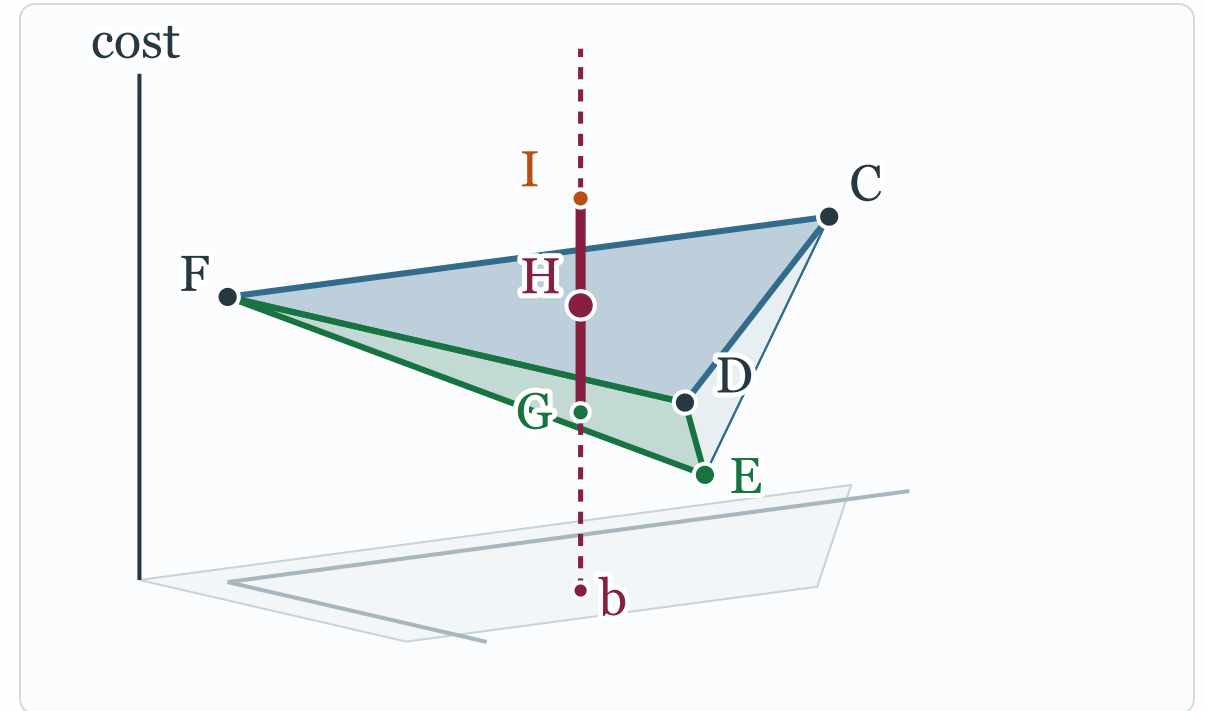


Adding E creates four possible boundary faces

The current triangle CDF together with E forms the tetrahedron CDEF.

Its four faces are CDF, CDE, CEF and DEF. The three new faces are candidates obtained by removing F, D or C.

In general an $(m + 1)$ -simplex has $m + 2$ faces: the old basic simplex plus $m + 1$ candidate replacements.

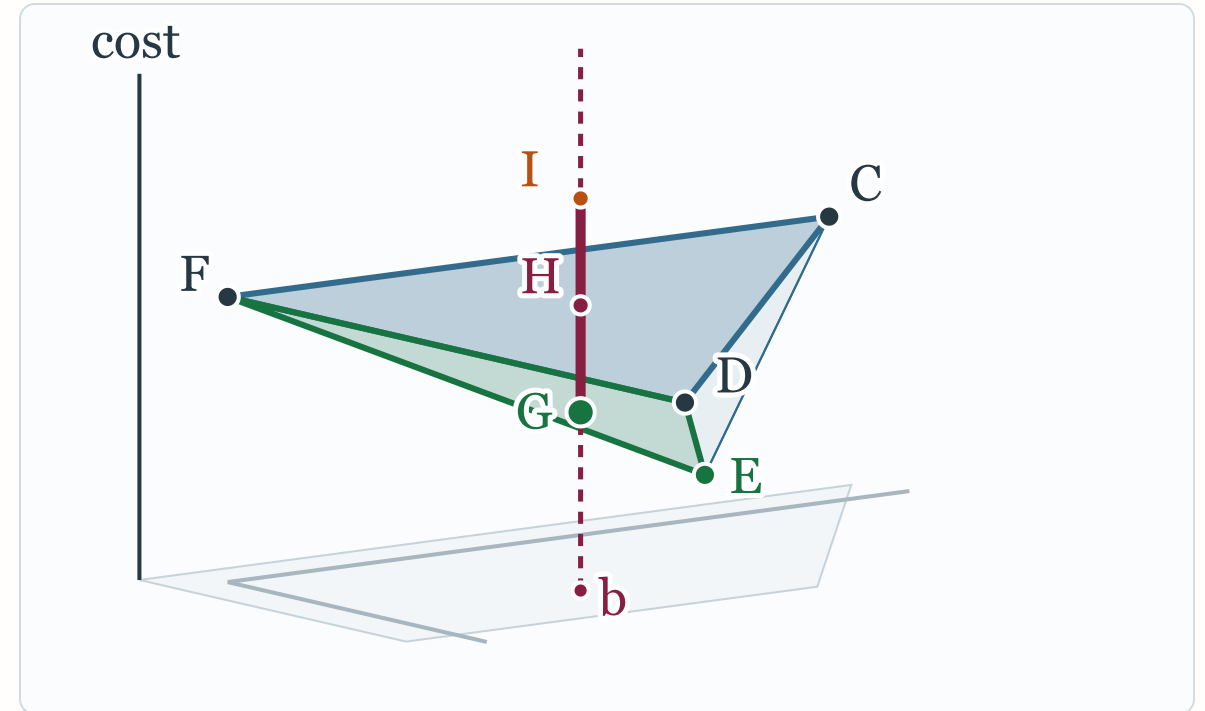


The requirement line selects the feasible new face

At H, the requirement line meets the old face CDF. Follow the line downward through the tetrahedron.

The next boundary intersection is G, on face DEF. That face contains E and excludes C: E enters and C leaves.

A candidate face that does not meet the requirement line cannot give nonnegative weights for this requirement.



The same face choice comes from the ratio test

Let $x_E = \theta$, starting from $x_C = x_D = x_F = 1/3$.
Preserving all three equalities gives

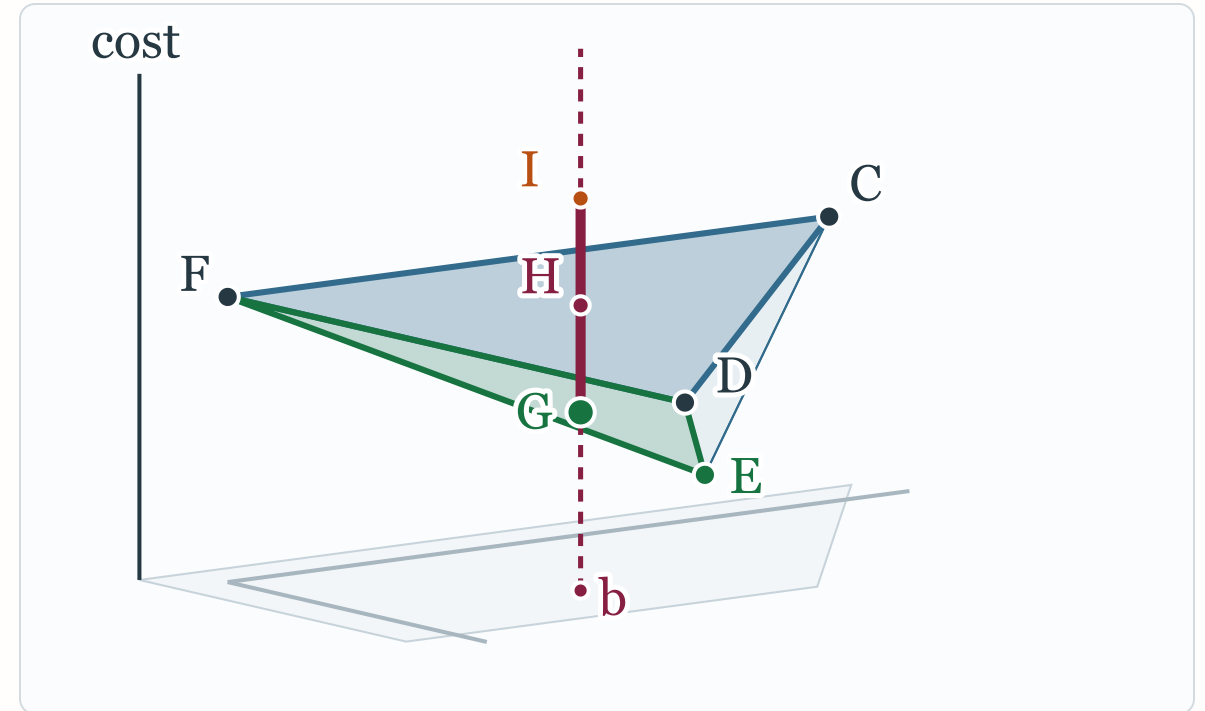
$$x_C = \frac{1}{3} - \frac{2}{3}\theta,$$

$$x_D = \frac{1}{3} - \frac{1}{6}\theta,$$

$$x_F = \frac{1}{3} - \frac{1}{6}\theta.$$

$$\theta^* = \min\{1/2, 2, 2\} = 1/2.$$

C reaches zero first, producing the face DEF.

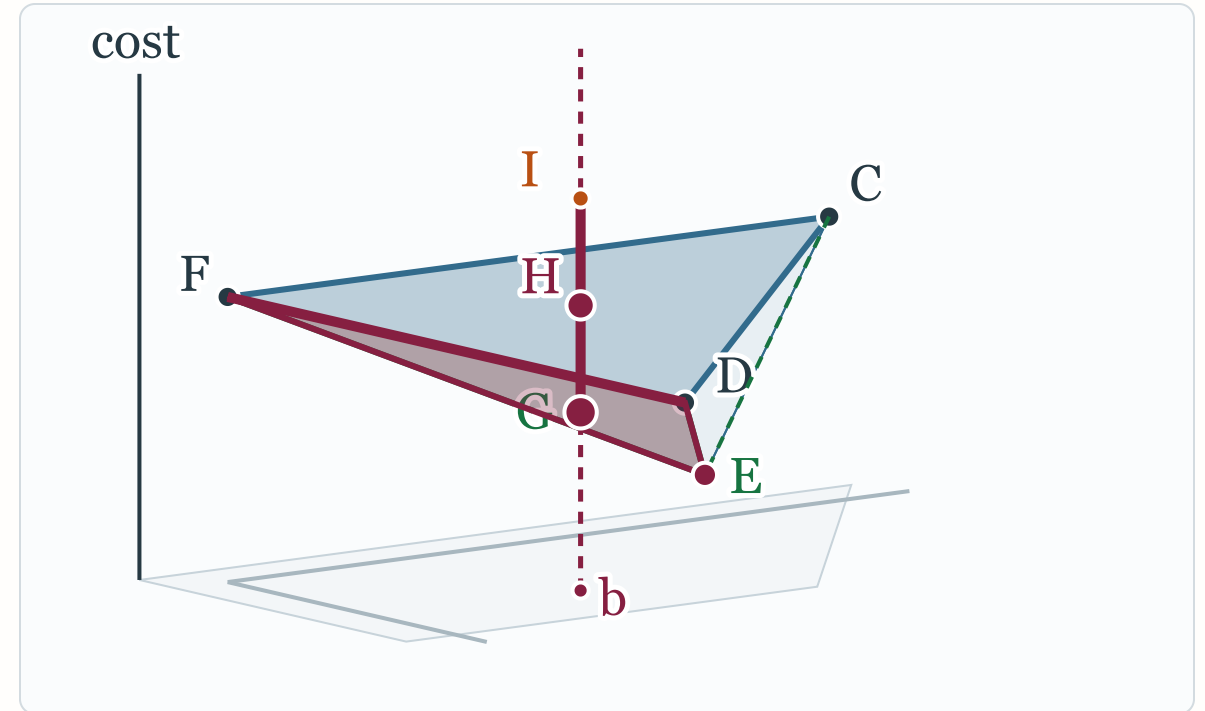


Pivot the triangle from CDF to DEF

The old triangle CDF and new triangle DEF share the hinge DF.

$$H = (1, 1, 4) \longrightarrow G = (1, 1, 5/2).$$

C leaves, E enters, and the cost decreases. The dashed corner path shows the physical analogy.



New triangle DEF; cost 2.50.

The final plane supports the entire convex hull

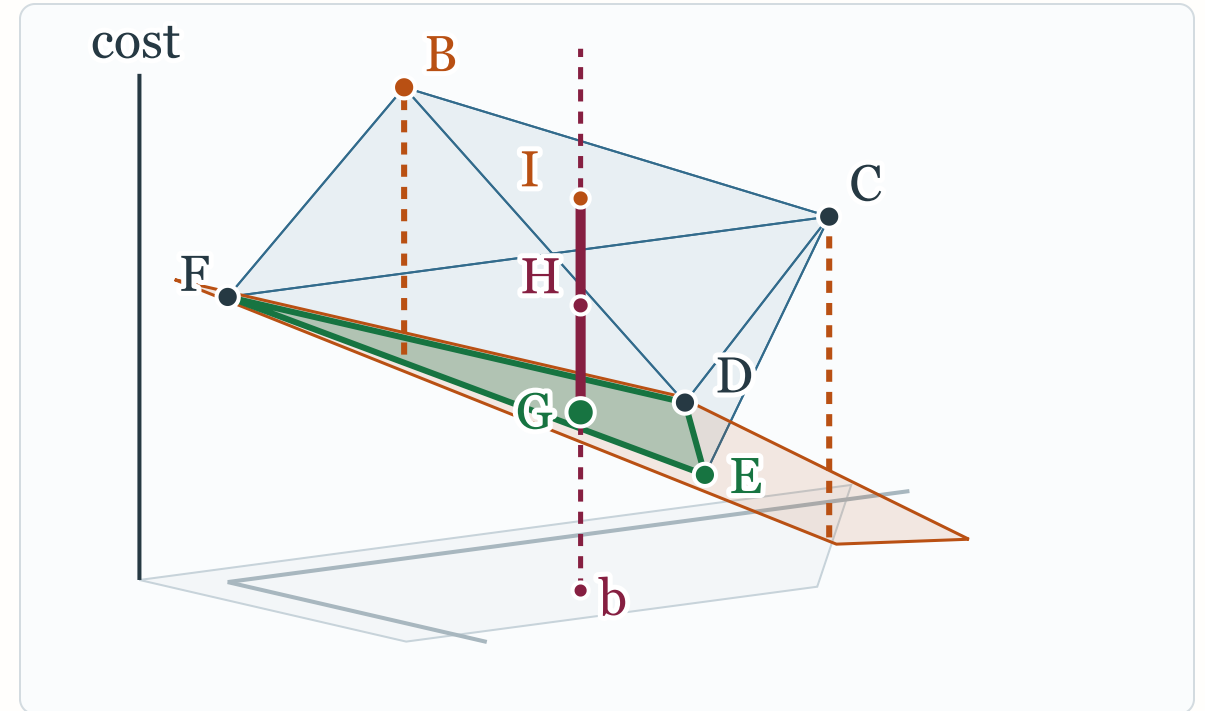
The plane through D, E and F is

$$z = 4 - \frac{3}{2}a_1.$$

Every lifted column lies on or above it. Therefore every convex combination satisfies $z \geq 4 - \frac{3}{2}a_1$.

At $b = (1, 1)$, this gives $z \geq 5/2$. The mixture G attains the bound, so it is optimal.

Self-study: verify the five signed gaps using the coordinate table.



Example 3.10: eight points, one horizontal coordinate

With $m = 1$, the basic simplex is a line segment. Use this constructed numerical realization of the eight-point picture:

j	1	2	3	4	5	6	7	8
A_j	5	1	-2	-3	3	4	2	-1
c_j	7	6	6	5	2	8	5	3

$$\min \sum_{j=1}^8 c_j x_j \quad \text{s.t.} \quad \sum_{j=1}^8 A_j x_j = 0, \quad \sum_{j=1}^8 x_j = 1, \quad x \geq 0.$$

We will choose the point with the most negative reduced cost: the largest downward vertical gap from the current dual line.

Initial basis 3, 6: point 5 has the largest downward gap

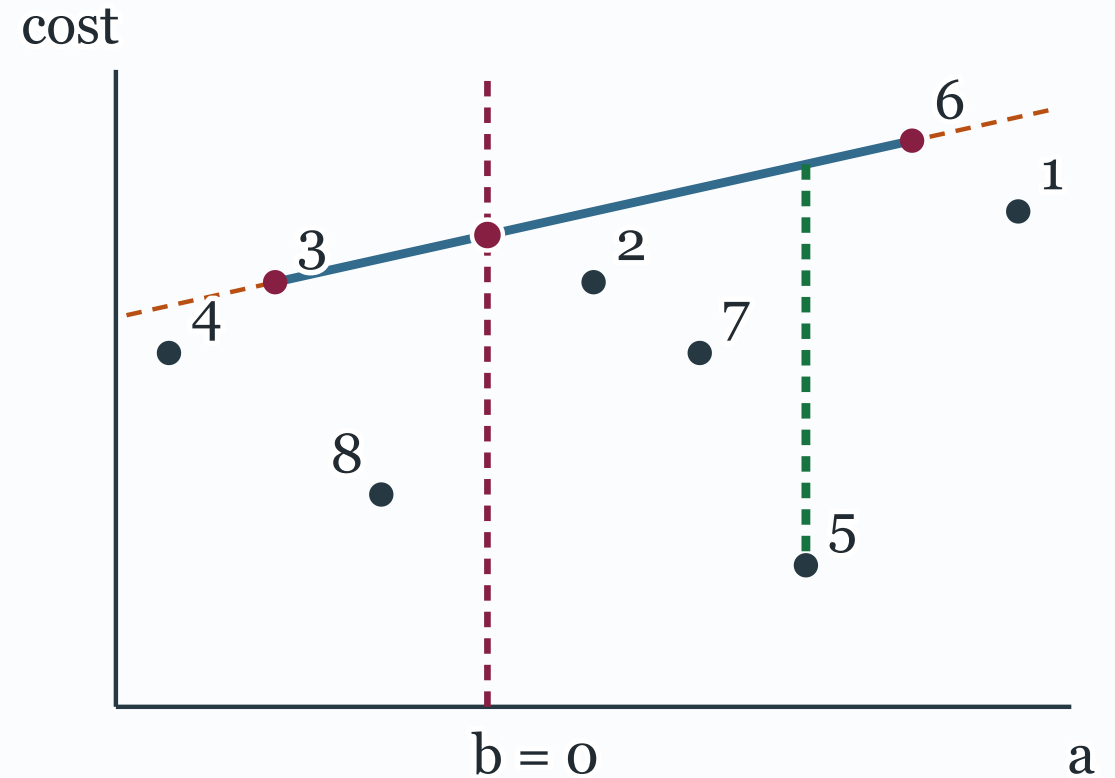
The line segment between points 3 and 6 crosses the requirement line.

$$x_3 = \frac{2}{3}, \quad x_6 = \frac{1}{3}, \quad z = \frac{20}{3}.$$

$$z = \frac{1}{3}a + \frac{20}{3}.$$

Point 5 has gap $2 - 23/3 = -17/3$, the most negative reduced cost.

Point 5 enters. The requirement lies between 3 and 5, so point 6 leaves.



Next basis 3, 5: point 8 is now the best candidate

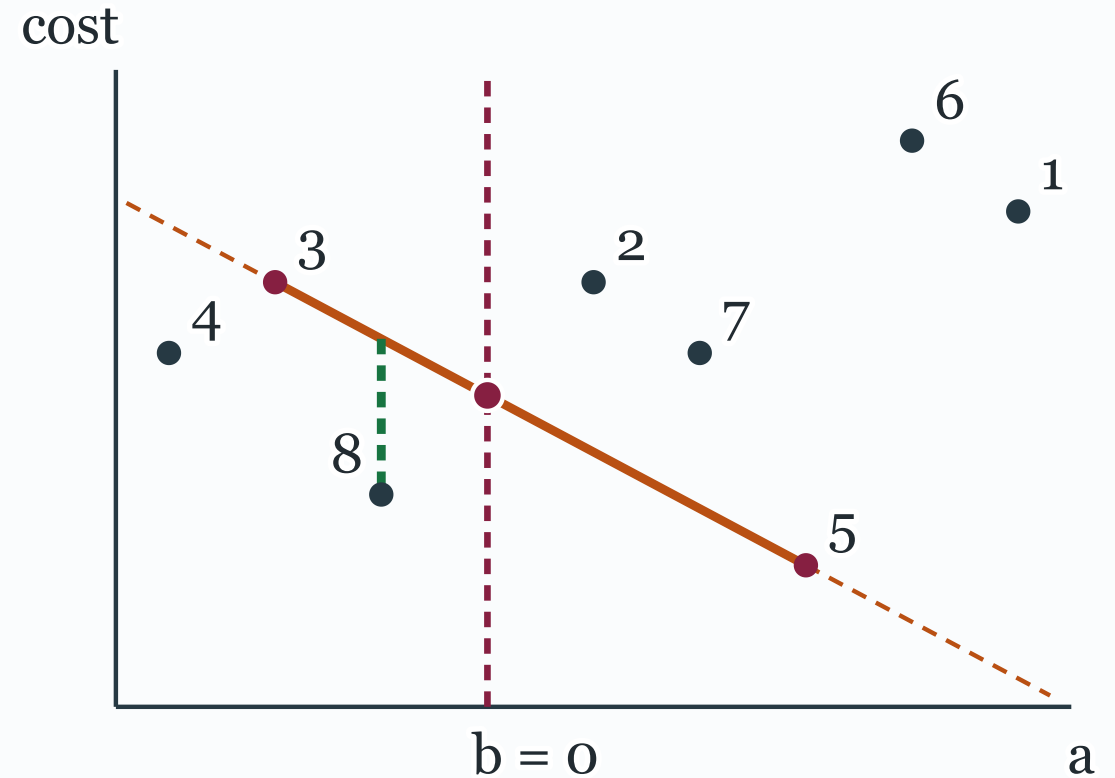
After the first pivot, the segment is 3, 5.

$$x_3 = \frac{3}{5}, \quad x_5 = \frac{2}{5}, \quad z = \frac{22}{5}.$$

$$z = -\frac{4}{5}a + \frac{22}{5}.$$

Now point 8 has the most negative gap: $3 - 26/5 = -11/5$.

Point 8 enters and point 3 leaves. The new segment must still cross the requirement line.



Final basis 5, 8: every point is on or above the line

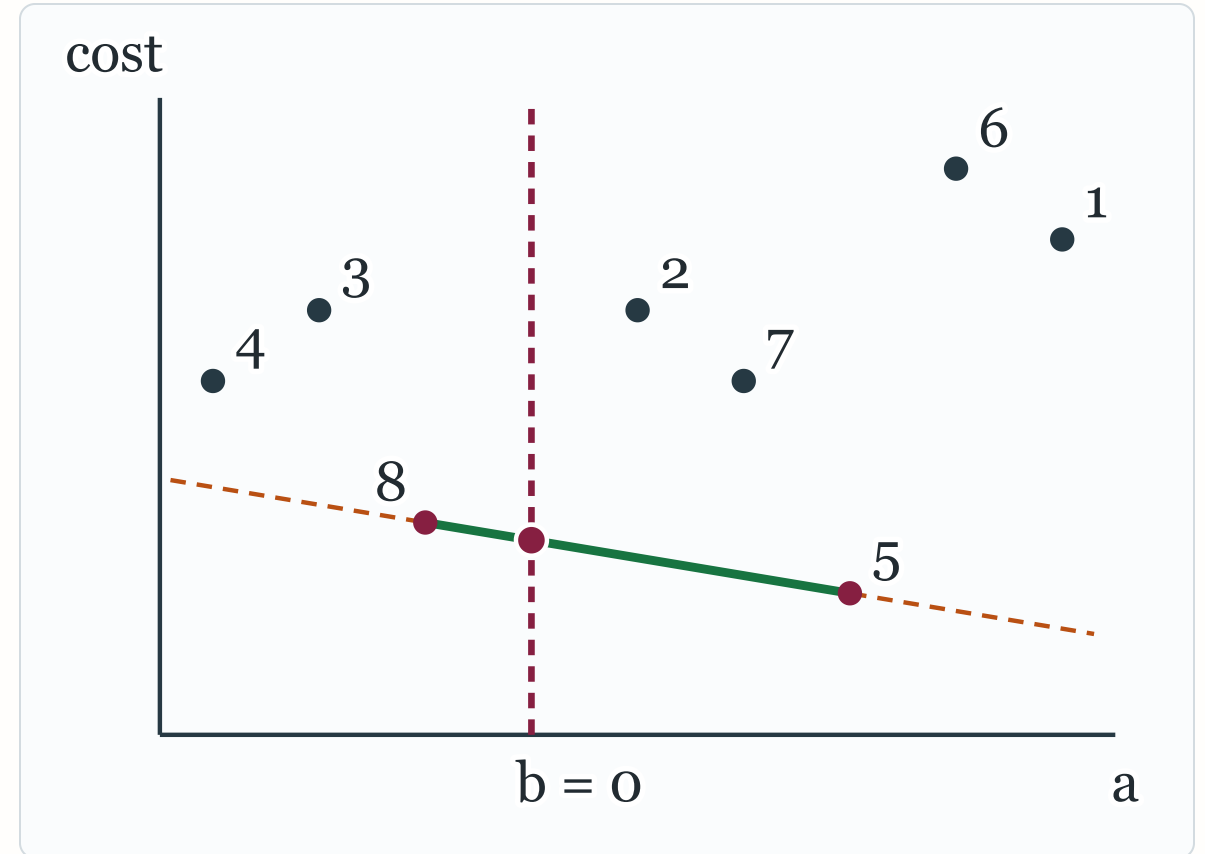
The final segment uses points 5 and 8.

$$x_5 = \frac{1}{4}, \quad x_8 = \frac{3}{4}, \quad z = \frac{11}{4}.$$

$$z = -\frac{1}{4}a + \frac{11}{4}.$$

All remaining points are above this line. Every nonbasic reduced cost is positive.

The current feasible mixture attains the lowest possible height at the requirement.



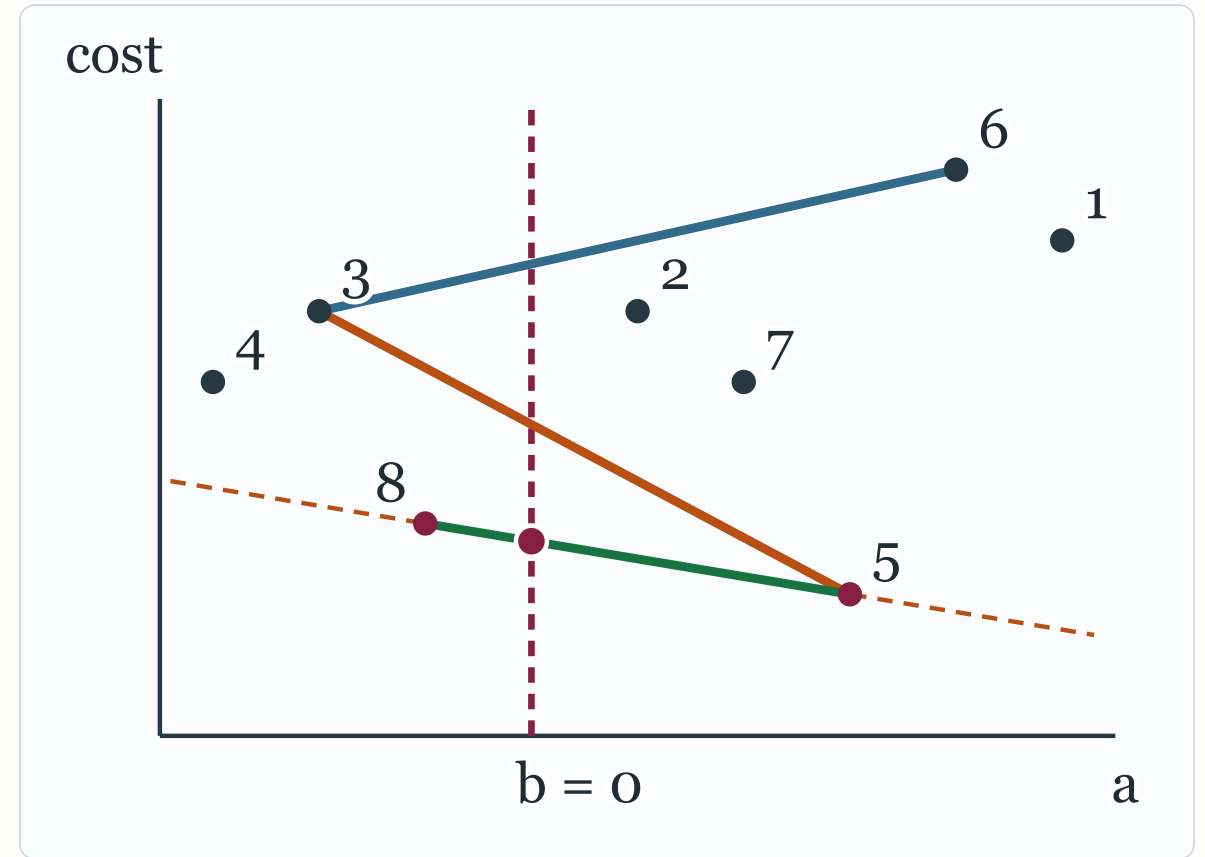
Read the entire path: 3, 6 → 3, 5 → 5, 8

Each pivot changes one endpoint of the basic segment while preserving its intersection with the requirement line.

$$\frac{20}{3} > \frac{22}{5} > \frac{11}{4}.$$

Self-study: recover the weights by solving the two equalities for each pair. Then calculate every reduced cost to confirm the entering choices.

The 2D segment rotates at a shared endpoint. The 3D triangle pivots around a shared edge. Both pictures describe the same basis exchange.



— initial: 3, 6 — next: 3, 5 — optimal: 5, 8

How much work does simplex require?

We now know how to start, pivot, and avoid cycling. Termination alone does not tell us how long the algorithm takes.

$$\text{total work} \approx \text{work per pivot} \times \text{number of pivots}.$$

We will examine both factors, then separate a deliberately difficult LP from typical behavior.

First factor: the work in one pivot

For standard form, let m be the number of independent equations and n the number of variables.

A full tableau has $O(mn)$ entries. One pivot updates them by row operations: $O(mn)$ arithmetic operations.

Revised simplex maintains and updates basis factors. Dense worst-case pricing and updates can also cost $O(mn)$, since $m \leq n$.

A fresh dense factorization costs $O(m^3)$. Sparsity matters; arithmetic counts also omit the bit length of exact numbers.

Bland's entering scan: compute reduced costs in increasing variable-index order; stop at the first negative one. If none is negative, check all nonbasic variables.

Why Lexicographic Comparison Needs More Than a Ratio

For basis B , write $\bar{b} = B^{-1}b$ and $\bar{A} = B^{-1}A$. Suppose x_j enters; consider a row with $\bar{a}_{ij} > 0$.

Ordinary test: compare one ratio. **Lex rule:** if those ratios tie, compare further entries in the fixed column order.

$$\frac{\bar{b}_i}{\bar{a}_{ij}} \quad \text{versus} \quad \frac{1}{\bar{a}_{ij}} \left[\bar{b}_i \mid \bar{a}_{i1}, \dots, \bar{a}_{in} \right].$$

Full tableau: store $[\bar{b} \mid \bar{A}]$. Read each candidate row and divide it by its positive entering-column coefficient.

Revised simplex: store basis factors for solving linear systems, instead of the full tableau. Obtain the needed normalized row for lex comparison by linear solves, without explicitly forming the inverse.

Second factor: how many pivots?

Many practical LPs need relatively few pivots. The reference describes counts roughly proportional to problem dimensions as an empirical observation.

That observation is not a bound for every LP. The feasible region can have exponentially many vertices.

To show slow behavior, we need more than many vertices: an **improving edge path** that a pivoting rule actually follows.

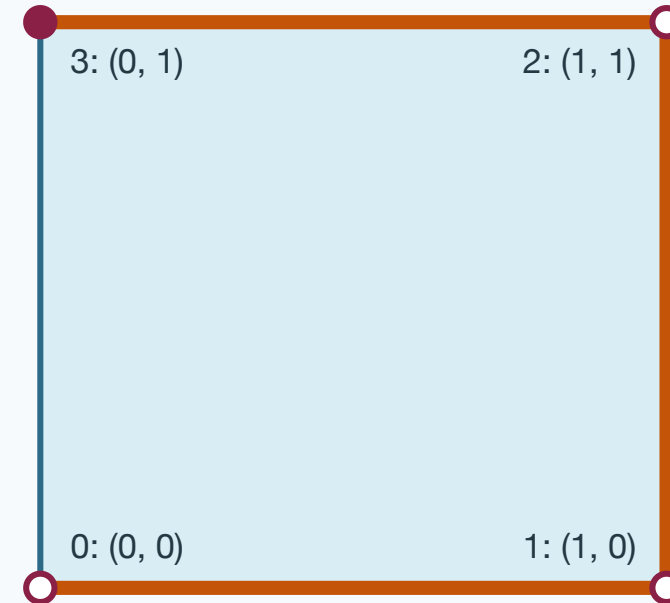
A cube already has exponentially many corners

Consider the unit cube:

$$0 \leq x_i \leq 1, \quad i = 1, \dots, n.$$

At a vertex each coordinate is either 0 or 1. There are 2^n binary choices.

A spanning path visits every vertex exactly once using edges.



minimize $f = -x_2$

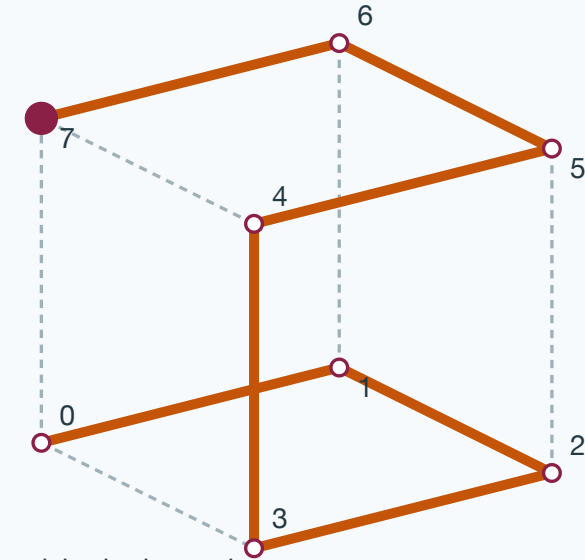
In two dimensions: 4 vertices and a path of 3 edges.

Build the path one dimension at a time

1. Follow the square path on $x_3 = 0$.
2. Cross one edge to $x_3 = 1$.
3. Follow the same square path in reverse.

$$L_n = 2L_{n-1} + 1 = 2^n - 1.$$

The same construction works recursively in every dimension.



orange: visited edges · ring: current vertex

Vertex 7 of 7; $x = (0, 0, 1)$; $f = -1$.

Why the ordinary cube is not enough

Choose the minimization objective $f(x) = -x_n$.

On the cube, half the vertices have $f = 0$, and half have $f = -1$.

Our spanning path has many edges along which the objective stays constant. It does not yet demonstrate a long sequence of *strictly improving* pivots.

Next we tilt the bounds, so movement in an earlier coordinate changes the possible value of the last coordinate.

Tilt the cube while keeping its combinatorics

Choose $0 < \epsilon < \frac{1}{2}$ and solve

$$\begin{aligned} \min \quad & -x_n \\ \text{s.t.} \quad & \epsilon \leq x_1 \leq 1, \\ & \epsilon x_{i-1} \leq x_i \leq 1 - \epsilon x_{i-1}, \quad i = 2, \dots, n. \end{aligned}$$

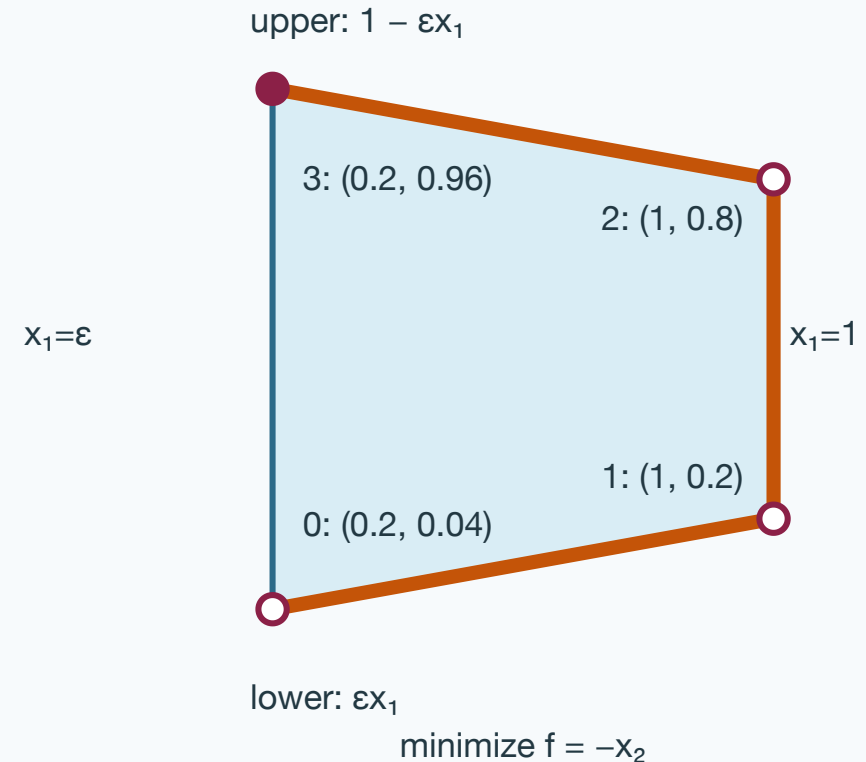
Each coordinate still has a lower and an upper bound. Their dependence on the previous coordinate makes the long path strictly improve the objective.

See the improvement in two dimensions

Set $\epsilon = 0.2$. Start at the vertex labeled 0 and follow the orange path.

vertex	0	1	2	3
x_2	.04	.20	.80	.96
$f = -x_2$	-.04	-.20	-.80	-.96

All three edge moves strictly decrease the objective. The fourth vertex is optimal.

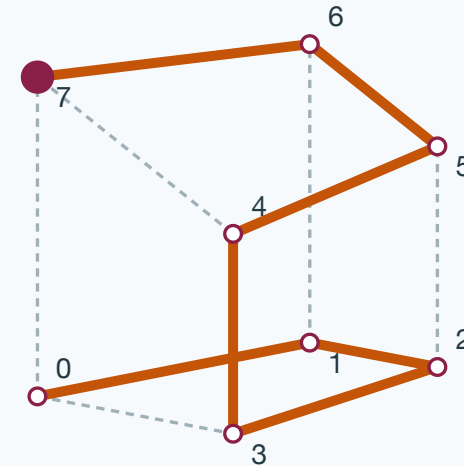


The same construction gives seven improving edges

Use the same inequalities with $n = 3$ and $\epsilon = 0.2$.

The figure shows the complete path from vertex 0 to vertex 7. The objective $f = -x_3$ decreases at every step.

The path visits all 8 vertices. In dimension n , it visits all 2^n vertices.



orange: visited edges · ring: current vertex

Vertex 7 of 7; $x = (0.2, 0.04, 0.992)$; $f = -0.992$.

Theorem 3.5: an exponential pivot path exists

For the perturbed cube just defined:

1. The feasible set has 2^n vertices.
2. They can be ordered so consecutive vertices share an edge and the cost strictly decreases.
3. A pivoting rule that follows this order takes $2^n - 1$ pivots.

This is an existence statement about a bad path and a rule that follows it. It does not say every rule follows that path.

Why there are exactly 2^n vertices

Choose a lower or upper bound for each coordinate, starting with x_1 .

The resulting triangular system has diagonal coefficients 1, so it defines a unique point.

Each lower bound is strictly below its upper bound: $\epsilon x_{i-1} < 1 - \epsilon x_{i-1}$, since $x_{i-1} \leq 1$ and $\epsilon < \frac{1}{2}$.

At most one bound per coordinate can be tight. A vertex needs n independent tight bounds, hence exactly one per coordinate. Every binary choice gives one distinct vertex.

Why reversing the upper path preserves improvement

On the lower face, $x_n = \epsilon x_{n-1}$: increasing x_{n-1} increases x_n .

Cross to the upper face, $x_n = 1 - \epsilon x_{n-1}$. This increases x_n , because $\epsilon x_{n-1} < 1 - \epsilon x_{n-1}$.

Then traverse the previous path backward. Now x_{n-1} decreases, so $1 - \epsilon x_{n-1}$ increases.

Consecutive vertices share $n - 1$ independent tight bounds. They are adjacent, and $-x_n$ strictly decreases throughout.

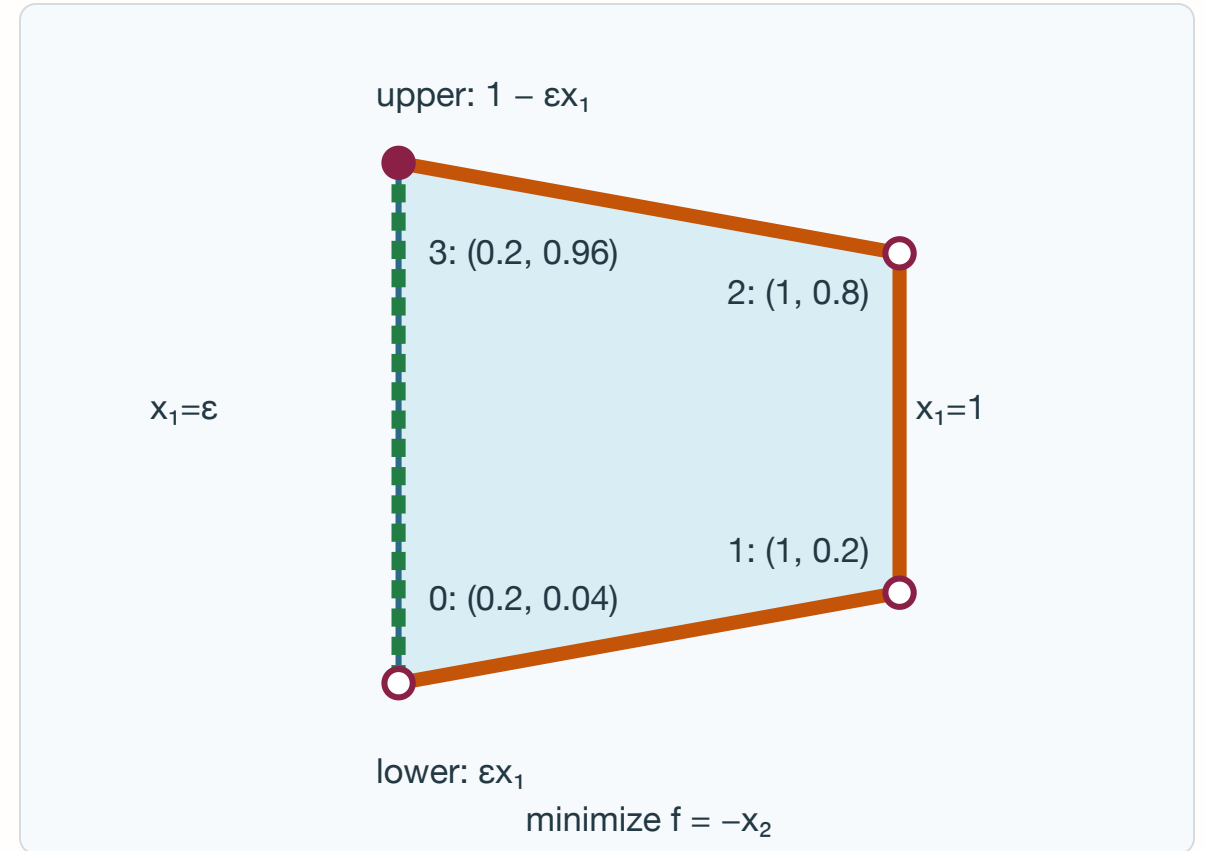
A different pivot choice can take a shortcut

The first and last vertices of this spanning path are adjacent.

Orange: follow the three-edge detour. Green: move directly to the optimum in one pivot.

The same first-to-last adjacency holds in the higher-dimensional construction.

Choosing a rule matters. A long available path does not force every algorithm to use it.



Can a pivoting rule always find a short route?

Exponential examples are known for many individual pivoting rules. That does not rule out every possible choice of rule.

The general search for a polynomial-time pivot rule remains a central question. A lower bound for one rule is not a lower bound for all rules.

We next separate two questions: **does a short path exist?** And **can an algorithm find an improving one efficiently?**

See [Disser and Mosis \(2025\)](#) for the scope of this open question.

Measure the shortest paths in the vertex graph

For a nonempty pointed polyhedron with vertices v, w , let $d(v, w)$ be the **minimum number of edges** in a path joining them.

$$D(P) = \max_{v, w \text{ vertices of } P} d(v, w).$$

This maximum of shortest-path distances is the **diameter** of the polyhedron's vertex-edge graph.

Distance counts edges, not Euclidean length. The paths in this definition need not improve an objective.

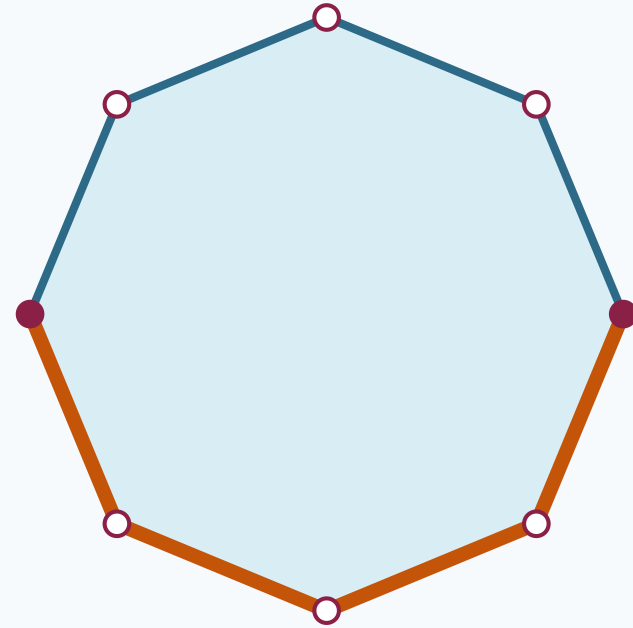
In two dimensions, a bounded polygon is a cycle

Here m counts inequality constraints and n is the ambient dimension. A bounded full-dimensional polygon has at most m edges.

Between two vertices, choose the shorter route around the boundary.

$$D(P) \leq \left\lfloor \frac{m}{2} \right\rfloor.$$

The illustrated octagon attains 4.



cycle: $D = 4 = \text{floor}(m / 2) \cdot m = 8$

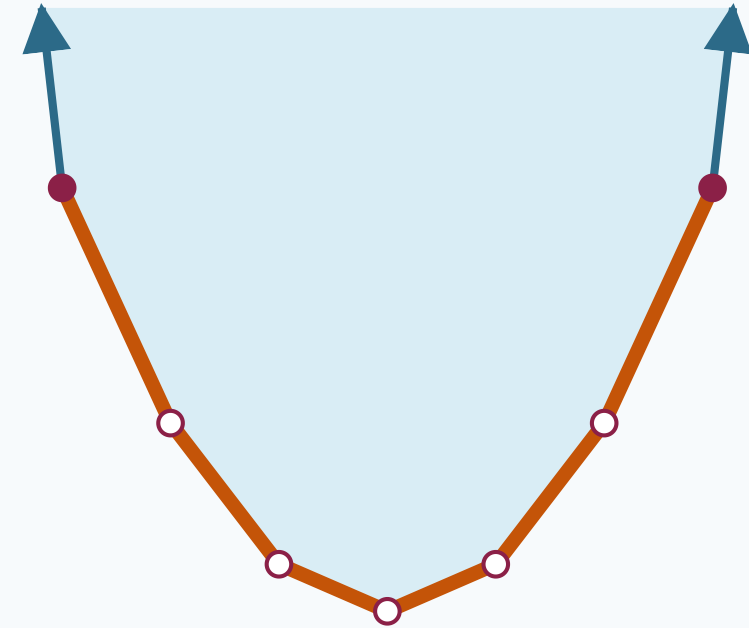
An unbounded polygon can have a longer chain

For a pointed, full-dimensional unbounded polygon, two boundary edges are rays. The finite vertices form a chain.

$$D(P) \leq m - 2.$$

With $m = 8$, the illustrated chain has seven vertices and six finite edges.

Thus the worst case over bounded and unbounded polygons is $\Delta(2, m) = m - 2$ for $m \geq 3$.



chain: $D = 6 = m - 2 \cdot m = 8$

Why diameter constrains every edge-following method

Suppose $d(v, w) = D(P)$. Start at v , and choose an objective for which w is the unique optimal vertex.

Every edge-following route from v to w needs at least $D(P)$ edge moves.

Define $\Delta(n, m)$ as the largest diameter among pointed n -dimensional polyhedra described by m inequalities.

A polynomial bound on diameter is necessary for a universally short edge path. It would not by itself tell us how to choose the pivots.

The Hirsch conjecture proposed a linear bound

The proposed bound was

$$\Delta(n, m) \leq m - n.$$

Klee and Walkup (1967) gave an unbounded-polyhedron counterexample.

Santos announced a bounded-polytope counterexample in 2010, published in 2012. The original Hirsch bound is therefore false even for polytopes.

[Santos, A counterexample to the Hirsch conjecture.](#)

A weaker question asks for a polynomial bound

The **polynomial Hirsch conjecture** asks whether $\Delta(n, m)$ is bounded by some polynomial in n and m .

A classical bound is

$$\Delta(n, m) \leq m^{1+\log_2 n} = (2n)^{\log_2 m}.$$

The exponent depends on the dimension. This is a quasipolynomial bound, not a polynomial bound jointly in both parameters.

The false linear conjecture and the weaker polynomial question must be kept separate.

A short path need not give a fast algorithm

Suppose a polynomial diameter bound were proved. Would that make simplex a polynomial-time algorithm?

No. Diameter promises some short edge path. It does not promise an improving path, identify the path, or bound the work needed to find it.

Degenerate pivots may also change the basis without traversing an edge to a new vertex. Their number needs its own control.

Worst-case behavior does not describe every instance

A worst-case result asks how badly a method can behave. An average-case result asks how it behaves under a specified distribution of inputs.

Many LPs encountered in practice require modest pivot counts. The reference describes roughly linear observed counts; these are observations, not universal theorems.

To turn “usually fast” into a mathematical claim, we must say what *usually* means.

Define the random experiment before taking an average

1. Fix the problem dimensions and choose a probability distribution over LP data.
2. Specify the algorithm, its initialization, and its pivoting rule.
3. Let T count the iterations and study $\mathbb{E}[T]$.

There is no single natural uniform distribution over all LPs. A result for one random model need not describe a different application.

One model: randomly orient fixed inequalities

Fix $c, a_1, \dots, a_m \in \mathbb{R}^n$ and $b_1, \dots, b_m$. Independently choose, with equal probability,

$$a_i^\top x \leq b_i \quad \text{or} \quad a_i^\top x \geq b_i.$$

There are 2^m sign choices. Suppose $L > 0$ produce feasible LPs.

The reference's Haimovich (1983) result bounds the average by $n/2$ pivots over those L feasible problems, for the specially specified pivoting procedure.

The distribution and procedure are essential assumptions; this is not a bound for arbitrary simplex implementations.

Smoothed analysis: perturb an arbitrary starting instance

Average-case analysis randomizes the whole input. **Smoothed analysis** starts with arbitrary data and adds small random perturbations.

For an appropriate shadow-vertex simplex algorithm, Gaussian perturbation models give polynomial expected complexity in the problem dimensions and $1/\sigma$, where σ measures the noise scale.

This helps explain why carefully constructed difficult instances can be fragile. It does not give every pivoting rule a polynomial worst-case guarantee.

Spielman and Teng, Smoothed Analysis (2004).

Four questions now form one complete story

- 1. Can pivots repeat forever?** Lexicographic and Bland rules prevent cycling.
- 2. How do we start?** Phase I finds a feasible basis or proves infeasibility.
- 3. What is a pivot geometrically?** Replace one basic point. A positive improving step lowers the intersection; a degenerate pivot can leave it unchanged.
- 4. How long can this take?** Count work per pivot and pivots; distinguish worst-case, average-case, and smoothed claims.

References and Paths for Independent Study

Practice the complete method

Construct Phase I, interpret its optimum, clean up the basis, and restore the original objective. Explain a pivot using both the dictionary and the column picture.

Textbook: Bertsimas and Tsitsiklis, *Introduction to Linear Optimization*, §§3.5–3.7.

Review: [Simplex II: Cycling and Anticycling](#) develops the finite-termination rules used in both phases.

Further reading: [Santos: the Hirsch counterexample](#); [Spielman and Teng: smoothed analysis](#); [Disser and Mosis: the pivot-rule complexity question](#).

Distinguish the existence of a short edge path, an improving path, and an algorithm that finds such a path efficiently.